AP CALCULUS BC · EXAMINATION REPOSITORY

Unit 7
Differential Equations

100-Question Priority-Starred Master Set

- Questions 1–90: Multiple Choice (Full Priority Ratings · 4 Questions Per Page)
- Questions 91–100: Free Response with Official AP Scoring Rubrics (9 Pts Each)
- Integrated AP Slope Field Diagrams, Logistic S-Curves & Euler's Method Data Tables
- Complete 90-Question Answer Key & Step-by-Step Bulleted Solutions with Distractor Analysis

OFFICIAL AP CED ALIGNED · MEDIUM LEVEL · RIGOROUSLY VERIFIED

Examination Structure

This comprehensive paper contains 100 continuously numbered questions strictly covering Unit 7 (Differential Equations), incorporating official AP-style vector slope fields, tables of values, and logistical diagrams to test conceptual and graphical calculus mastery.

⭐ Priority Rating System

★★★★★ VERY IMPORTANT	High-frequency core exam topics: Separation of Variables with Initial Conditions, Slope Fields Reasoning, Euler's Method (BC), and Logistic Growth Models (BC).
★★★★ MANDATORY	Essential analytical skills: Solution domains of differential equations, Exponential Growth/Decay models, Newton's Law of Cooling, and Second Derivative / Concavity tests.
★★★ IMPORTANT	Foundational practice: Differential equation modeling, proportional rates, verifying solutions via substitution, and slope field matchings.

- **Questions 1–90:** Multiple Choice, four choices, exactly one correct answer (Exactly 4 MCQs per page).
- **Questions 91–100:** Free Response Questions, 9 points each, with authentic AP multi-part rubrics.
- **Visual Elements:** Crisp vector SVG slope field diagrams, logistic growth curves, mixing tank schematics, and numerical step tables.
- **Calculator Policy:** No calculator required (Closed-form analytical evaluation).
- **Answer Section:** Complete Answer Key grid, followed by step-by-step bulleted solutions and distractor breakdowns.

Topic Coverage (Unit 7.1 – 7.9)

7.1 Modeling Situations with Differential Equations	7.2 Verifying Solutions for Differential Equations
7.3 Sketching Slope Fields	7.4 Reasoning Using Slope Fields
7.5 BC: Euler's Method Approximations	7.6 Finding General Solutions (Separation of Variables)
7.7 Particular Solutions & Solution Domains	7.8 Exponential Models & Newton's Cooling
7.9 BC: Logistic Models & Carrying Capacity	Synthesis: Second Derivatives & Mixing Rates

Essential Tip: In slope field problems, look for invariant lines (where slopes are constant along horizontal lines $y = c$ or vertical lines $x = k$) and identify points where $dy/dx = 0$ (horizontal tangent segments).

ITEM 01

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

The rate of change of the volume V of a melting snowball with respect to time t is directly proportional to its surface area S. Which of the following differential equations models this relationship, where k is a positive constant?

A $dV/dt = k\,S$
B $dV/dt = -k\,S$
C $dV/dt = -k / S$
D $dS/dt = -k\,V$

ITEM 02

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

A population P of bacteria grows at a rate directly proportional to the square root of the population size at any time t. If k is a positive constant, which differential equation models this situation?

A $dP/dt = k\,P^2$
B $dP/dt = k / \sqrt{P}$
C $dP/dt = k\sqrt{P}$
D $dP/dt = \sqrt{k} \cdot P$

ITEM 03

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

According to Newton's Law of Cooling, the rate of change of the temperature T of an object is proportional to the difference between its temperature and the ambient room temperature M. If $T > M$ and $k > 0$, which differential equation correctly represents this process?

A $dT/dt = -k\,(T - M)$
B $dT/dt = k\,(T + M)$
C $dT/dt = -k\,(M - T)$
D $dT/dt = -k\,T \cdot M$

ITEM 04

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

Which of the following functions is a solution to the differential equation $dy/dx = 3y$?

A $y = 3e^x$
B $y = e^{3x}$
C $y = x^3$
D $y = \sin(3x)$

ITEM 05

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

Which of the following functions is a solution to the differential equation $y'' + 4y = 0$?

A $y = e^{2x}$
B $y = e^{-2x}$
C $y = \cos(2x)$
D $y = \cos(4x)$

ITEM 06

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For what value of the constant k is $y = x^2 + k/x$ a solution to the differential equation $x \cdot (dy/dx) + y = 3x^2$ for $x > 0$?

A k = 0 only
B k = 3
C Any real number k
D No value of k

ITEM 07

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

If $y = C\,e^{-2x} + 3$ is a family of solutions to a differential equation, which of the following could be that differential equation?

A $dy/dx = -2y + 6$
B $dy/dx = -2y - 6$
C $dy/dx = 2y - 6$
D $dy/dx = -2y + 3$

ITEM 08

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

A water tank drains such that the rate of change of the depth h of water (in feet) with respect to time t (in minutes) is inversely proportional to the square of the depth. If k is a positive constant, which differential equation models this relationship?

A $dh/dt = -k\,h^2$
B $dh/dt = -k / h^2$
C $dh/dt = k / h^2$
D $dh/dt = -k / \sqrt{h}$

ITEM 09

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following functions satisfies the initial value problem $dy/dx = 2x\,y$ with $y(0) = 5$?

A $y = 5e^{2x}$

B $y = e^{x^2} + 4$

C $y = 5e^{x^2}$

D $y = 5x^2 + 5$

ITEM 10

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The function $y = x \ln x$ is a particular solution to which of the following differential equations for $x > 0$?

A $x \cdot (dy/dx) - y = x$

B $x \cdot (dy/dx) + y = x$

C $dy/dx = \ln x$

D $x^2 \cdot (dy/dx) = y$

ITEM 11

EASY · SINGLE BEST CHOICE ★★★ MANDATORY

A slope field shows line segments that have the same slope along any horizontal line (constant y). Which of the following differential equations could produce this slope field?

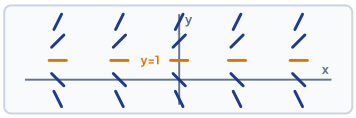


FIGURE 7.3: SLOPE FIELD WITH HORIZONTAL SEGMENTS ALONG Y = 1

A $dy/dx = 2x$

B $dy/dx = y - 1$

C $dy/dx = x + y$

D $dy/dx = x\,y$

ITEM 12

EASY · SINGLE BEST CHOICE ★★★ MANDATORY

A slope field has line segments with identical slope along any vertical line (constant x). Which differential equation could model this slope field?

A $dy/dx = y^2$

B $dy/dx = x / y$

C $dy/dx = x + y$

D $dy/dx = x - 2$

ITEM 13

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For the differential equation $dy/dx = x - 2y$, along which line in the xy-plane are all tangent line segments horizontal?

A $y = x/2$
B $y = 2x$
C $y = -x/2$
D $x = 0$

ITEM 14

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Consider the differential equation $dy/dx = x^2 y$. In which quadrant(s) of the xy-plane are all the slope segments strictly negative?

A Quadrant II only
B Quadrant IV only
C Quadrants III and IV only
D Quadrants II and III only

ITEM 15

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $y = f(x)$ be the particular solution to the differential equation $dy/dx = y - x$ with the initial condition $f(1) = 1$. What is the value of $f''(1)$?

A 0
B −1
C 1
D 2

ITEM 16

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $y = f(x)$ be the particular solution to $dy/dx = x\,y$ with $f(1) = 2$. Is the graph of f concave up or concave down at $x = 1$?

A Concave up, because $f''(1) = 4 > 0$
B Concave down, because $f''(1) = -4 < 0$
C Linear, because $f''(1) = 0$
D Cannot be determined without finding $f(x)$

ITEM 17

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Let $y = f(x)$ be the particular solution to $dy/dx = x + 2y$ with $f(0) = 3$. What is an equation of the line tangent to the graph of f at $x = 0$?

A $y = 3x + 6$

B $y = 2x + 3$

C $y = x + 3$

D $y = 6x + 3$

ITEM 18

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For which of the following differential equations are all tangent segments horizontal along the ellipse $2x^2 + y^2 = 4$?

A $dy/dx = 2x^2 + y^2$

B $dy/dx = 4 - 2x^2$

C $dy/dx = 2x^2 + y^2 - 4$

D $dy/dx = y^2 - 2x^2$

ITEM 19

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $y = f(x)$ be a particular solution to $dy/dx = (y - 4)(x^2 + 1)$. If $f(0) = 4$, what is $f(x)$ for all real x?

A $f(x) = 4$

B $f(x) = 4e^x$

C $f(x) = x^2 + 4$

D $f(x) = 4e^{x^3/3 + x}$

ITEM 20

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A slope field for the differential equation $dy/dx = (x - 3)(y + 2)$ is given. If a particular solution passes through the point $(3, 1)$, does this solution have a relative minimum, relative maximum, or neither at $x = 3$?

A Relative maximum

B Relative minimum

C Point of inflection with horizontal tangent

D Neither, because dy/dx is undefined at $x = 3$

ITEM 21

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let y = f(x) be the particular solution to the differential equation dy/dx = x + y with initial condition f(0) = 1. Using Euler's method with 2 steps of equal size Δx = 0.5, what is the approximation for f(1)?

Step n	x_n	y_n	dy/dx = x + y	Δy = (dy/dx)(0.5)
0	0.0	1.00	1.00	0.50
1	0.5	1.50	2.00	1.00
2	1.0	**2.50**	—	—

A 2.00 **B** 2.25

C 2.50 **D** 2.75

ITEM 22

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let y = f(x) be the particular solution to dy/dx = 2xy with f(0) = 1. Using Euler's method starting at x = 0 with step size Δx = 0.2, what is the approximation for f(0.4)?

Step n	x_n	y_n	dy/dx = 2xy	Δy = (dy/dx)(0.2)
0	0.0	1.00	0.00	0.00
1	0.2	1.00	0.40	0.08
2	0.4	**1.08**	—	—

A 1.00 **B** 1.08

C 1.16 **D** 1.20

ITEM 23

EASY · SINGLE BEST CHOICE ★★★ MANDATORY

If dy/dx = 3x − y with f(1) = 2, what is the Euler approximation for f(1.2) using a single step of size Δx = 0.2?

A 2.0 **B** 2.4

C 2.6 **D** 2.2

ITEM 24

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let y = f(x) be the solution to dy/dx = 2x − y with initial condition f(2) = 3. Using Euler's method with 2 steps of size Δx = −0.5, what is the approximation for f(1)?

A 2.25 **B** 2.50

C 2.00 **D** 1.75

ITEM 25

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particular solution $y = f(x)$ to $dy/dx = x^2 + y$ satisfies $f(0) = 1$. An approximation for $f(0.5)$ is computed using Euler's method with 2 equal steps. Is this Euler approximation an overestimate or an underestimate of the true value of $f(0.5)$?

A Overestimate, because $f''(x) > 0$ on $[0, 0.5]$

B Overestimate, because $f'(x) > 0$ on $[0, 0.5]$

C Underestimate, because $f''(x) > 0$ on $[0, 0.5]$

D Underestimate, because $f'(x) > 0$ on $[0, 0.5]$

ITEM 26

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Consider $dy/dx = y^2 - x$ with initial condition $y(1) = 2$. Use Euler's method with 2 equal steps of $\Delta x = 0.5$ to approximate $y(2)$.

A 5.375

B 7.250

C 10.750

D 8.875

ITEM 27

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The table below shows the execution of Euler's method with step size $\Delta x = 0.5$ to approximate $y(1)$ for a differential equation $dy/dx = f(x, y)$:

x: 0, 0.5, 1.0
y: 2, 3.5, k
dy/dx: 3, 4, —

What is the value of k?

A 5.0

B 5.5

C 6.0

D 7.5

ITEM 28

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

If $dy/dx = 2x$ with $f(0) = 0$, what is the Euler approximation for $f(0.3)$ using 3 steps of equal size $\Delta x = 0.1$?

A 0.06

B 0.09

C 0.04

D 0.08

ITEM 29

EASY · SINGLE BEST CHOICE ★★★ MANDATORY

Let $dy/dx = 4 - y$ with $f(0) = 2$. Using Euler's method with step size $\Delta x = 1$, what is the approximation for $f(2)$?

A 2
B 3
C 4
D 6

ITEM 30

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

When using Euler's method to approximate the value of a particular solution at a fixed point $x = b$, what happens as the step size Δx approaches 0 (with number of steps $n \to \infty$)?

A The approximation diverges to infinity
B The approximation remains constant regardless of step size
C The approximation becomes undefined due to floating point error
D The approximation converges to the exact value of the particular solution

ITEM 31

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the general solution to the differential equation $dy/dx = 2x\,y$ for $y \neq 0$?

A $y = e^{x^2} + C$
B $y = x^2 y + C$
C $y = C\,e^{x^2}$
D $y = C\,e^{2x}$

ITEM 32

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

For $y > 0$, what is the general solution to the differential equation $dy/dx = x\sqrt{y}$?

A $y = (x^2/4 + C)^2$
B $y = (x^2/2 + C)^2$
C $y = x^4/16 + C$
D $y = (x^2/4)^2 + C$

ITEM 33

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $y \neq 0$, what is the general solution to the differential equation $dy/dx = -2x\,y^2$?

A $y = -1/(x^2 + C)$
B $y = e^{-x^2} + C$
C $y = -x^2 + C$
D $y = 1/(x^2 + C)$

ITEM 34

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the general solution to the differential equation $dy/dx = e^{x+y}$?

A $y = \ln(e^x + C)$
B $y = -\ln(C - e^x)$
C $y = e^x + e^y + C$
D $y = -\ln(e^x + C)$

ITEM 35

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

For $y > 0$, what is the general solution to the differential equation $dy/dx = (\cos x)/(2y)$?

A $y^2 = \sin x + C$
B $y^2 = -\sin x + C$
C $y = \sin x + C$
D $2y^2 = \sin x + C$

ITEM 36

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the general solution to the differential equation $dy/dx = (x + 1)/y$ for $y \neq 0$?

A $y = x + 1 + C$
B $y^2 = (x + 1)^2 + C$
C $y^2 = x^2 + 2x + C$
D $y^2 = x^2/2 + x + C$

ITEM 37

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the general solution to the differential equation $dy/dx = (1 + x^2)\,y$ for $y > 0$?

A $y = e^{x + x^3/3} + C$

B $y = (x + x^3/3) + C$

C $y = C\,e^{2x}$

D $y = C\,e^{x + x^3/3}$

ITEM 38

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $y > 0$ and $-\pi/2 < x < \pi/2$, what is the general solution to $dy/dx = y \sec^2 x$?

A $y = \tan x + C$

B $y = C\,e^{\tan x}$

C $y = C\,e^{\sec x}$

D $y = \ln|\tan x| + C$

ITEM 39

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

When solving the differential equation $dy/dx = 4y$, a student writes:
$\ln|y| = 4x + C \Rightarrow y = e^{4x} + C$
Why is the student's second step mathematically invalid?

A Exponentiating both sides yields $e^{4x + C} = e^{4x} \cdot e^{C} = A\,e^{4x}$, which produces a multiplicative constant rather than an additive constant + C

B The derivative of $\ln|y|$ is $1/y^2$, not $1/y$

C The constant C must always equal 0 in exponential differential equations

D The differential equation $dy/dx = 4y$ cannot be solved by separation of variables

ITEM 40

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the general solution to the differential equation $dy/dx = 3(y - 1)$?

A $y = C\,e^{3x} - 1$

B $y = e^{3x + 1} + C$

C $y = 1 + 3x + C$

D $y = 1 + C\,e^{3x}$

ITEM 41

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Find the particular solution to the differential equation $dy/dx = 2x(y - 1)$ with the initial condition $y(0) = 3$.

A $y = 1 + 3e^{x^2}$

B $y = 1 + 2e^{x^2}$

C $y = 3e^{x^2}$

D $y = 1 + 2e^{2x}$

ITEM 42

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Find the particular solution $y = f(x)$ to the differential equation $dy/dx = xy^2$ satisfying the initial condition $f(0) = 1$.

A $y = 1/(1 - x^2)$

B $y = -2/(x^2 + 2)$

C $y = 1/(1 + x^2/2)$

D $y = 2/(2 - x^2)$

ITEM 43

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the particular solution to the differential equation $dy/dx = x/y$ that satisfies $y(1) = -3$?

A $y = \sqrt{(x^2 + 8)}$

B $y = -\sqrt{(x^2 + 10)}$

C $y = -\sqrt{(x^2 + 8)}$

D $y = -(x + 2)$

ITEM 44

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Find the particular solution $y = f(x)$ to the differential equation $dy/dx = (x^2 + 1)/y$ given that $y(0) = 2$.

A $y = \sqrt{((2/3)x^3 + 2x + 4)}$

B $y = \sqrt{(x^3/3 + x + 4)}$

C $y = \sqrt{((2/3)x^3 + 2x + 2)}$

D $y = (2/3)x^3 + 2x + 2$

ITEM 45

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Solve the initial value problem $dy/dx = y \cos x$ with the condition $y(0) = 5$ for $y > 0$.

A $y = e^{\sin x} + 4$

B $y = 5\,e^{\sin x}$

C $y = 5\,e^{-\sin x}$

D $y = 5 \sin x + 5$

ITEM 46

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Find the particular solution to the differential equation $dy/dx = 3x^2 / (2y + 1)$ with the initial condition $y(0) = 1$.

A $y = (-1 + \sqrt{(4x^3 + 5)}) / 2$

B $y = (1 + \sqrt{(4x^3 + 9)}) / 2$

C $y = (-1 + \sqrt{(4x^3 + 9)}) / 2$

D $y = \sqrt{(x^3 + 1)}$

ITEM 47

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

If $dy/dx = e^x / y^2$ and $y(0) = 1$, what is the particular solution for y?

A $y = (e^x)^{(1/3)}$

B $y = (3e^x - 2)^{(1/3)}$

C $y = (3e^x + 1)^{(1/3)}$

D $y = 3e^x - 2$

ITEM 48

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Solve the initial value problem $dy/dx = (2x + 1)\,y$ with initial condition $y(1) = e$ for $y > 0$.

A $y = e^{x^2 + x}$

B $y = e^{x^2 + x - 2}$

C $y = e \cdot e^{x^2 + x}$

D $y = e^{x^2 + x - 1}$

ITEM 49

HARD · SINGLE BEST CHOICE ★★★★ MANDATORY

Find the particular solution $y = f(x)$ to the differential equation $dy/dx = -y^2 / (x + 1)$ with $f(0) = 1$ for $x > -1$.

A $y = 1 / (\ln(x + 1) + 1)$

B $y = 1 / (1 - \ln(x + 1))$

C $y = -1 / (\ln(x + 1) + 1)$

D $y = e^{-x} + 1$

ITEM 50

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the particular solution to $dy/dx = (y - 4)^2$ with the initial condition $y(0) = 3$?

A $y = 4 + 1 / (x + 1)$

B $y = 4 - 1 / (x - 1)$

C $y = 4 - 1 / (x + 1)$

D $y = 3 / (1 - 3x)$

ITEM 51

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Consider the differential equation $dy/dx = 2x\,y^2$ with initial condition $y(0) = 1$. What is the particular solution $y = f(x)$ and its largest open interval of validity?

A $y = 1 / (1 + x^2)$ on $(-\infty, \infty)$

B $y = 1 / (1 - x^2)$ on $(-1, 1)$

C $y = 1 / (1 - x^2)$ on $(1, \infty)$

D $y = 1 / (x^2 - 1)$ on $(-1, 1)$

ITEM 52

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Find the particular solution to $dy/dx = 1 / (2y)$ with the initial condition $y(0) = -2$, including the domain of the solution.

A $y = -\sqrt{(x + 4)}$ on $(-4, \infty)$

B $y = \sqrt{(x + 4)}$ on $(-4, \infty)$

C $y = -\sqrt{(x + 4)}$ on $[-4, \infty)$

D $y = -\sqrt{(x + 2)}$ on $(-2, \infty)$

ITEM 53

HARD · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the particular solution to $dy/dx = y^2 \cos x$ with $y(0) = 1$, and on what interval containing $x = 0$ is it valid?

A $y = 1/(1 + \sin x)$ on $(-\pi/2, 3\pi/2)$

B $y = 1/(1 - \cos x)$ on $(0, 2\pi)$

C $y = 1/(1 - \sin x)$ on $(-3\pi/2, \pi/2)$

D $y = 1/(1 - \sin x)$ on $(-\pi/2, \pi/2)$

ITEM 54

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Find the particular solution $y = f(x)$ to $dy/dx = (x - 2)/y$ with $y(2) = -1$.

A $y = \sqrt{((x-2)^2 + 1)}$

B $y = -\sqrt{(x^2 - 4x + 3)}$

C $y = 1 - x$

D $y = -\sqrt{((x-2)^2 + 1)}$

ITEM 55

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

If $dy/dx = y^2/x$ with $y(1) = 2$ for $x > 0$, what is the domain of the particular solution?

A $x \in (0, \infty)$

B $x \in (0, e)$

C $x \in (0, \sqrt{e})$

D $x \in (\sqrt{e}, \infty)$

ITEM 56

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the particular solution to $dy/dx = 4x^3 y^2$ with $y(0) = 1/4$?

A $y = 1/(4 + x^4)$

B $y = 4/(1 - 4x^4)$

C $y = (x^4 + 4)/16$

D $y = 1/(4 - x^4)$

ITEM 57

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Find the particular solution $y = f(x)$ to $dy/dx = 1 / y$ given that $y(1) = 3$.

A $y = \sqrt{(2x + 8)}$

B $y = \sqrt{(x + 8)}$

C $y = \sqrt{(2x + 7)}$

D $y = 2x + 1$

ITEM 58

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Find the particular solution to $dy/dx = e^x / y$ with the initial condition $y(0) = -2$.

A $y = \sqrt{(2e^x + 2)}$

B $y = -\sqrt{(2e^x + 2)}$

C $y = -\sqrt{(2e^x + 4)}$

D $y = -\sqrt{(e^x + 3)}$

ITEM 59

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Find the particular solution to $dy/dx = (1 + y^2)\,x$ with $y(0) = 0$ and specify its largest open interval of definition.

A $y = \tan(x^2)$ on $(-\sqrt{(\pi/2)}, \sqrt{(\pi/2)})$

B $y = \tan(x^2/2)$ on $(-\sqrt{\pi}, \sqrt{\pi})$

C $y = \tan(x^2/2)$ on $(-\infty, \infty)$

D $y = \arctan(x^2/2)$ on $(-\sqrt{\pi}, \sqrt{\pi})$

ITEM 60

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Find the particular solution to $dy/dx = 2x / (3y^2)$ with initial condition $y(1) = 2$.

A $y = (x^2 + 8)^{(1/3)}$

B $y = (2x^2 + 6)^{(1/3)}$

C $y = (x^2 + 7)^3$

D $y = (x^2 + 7)^{(1/3)}$

ITEM 61

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A population of organisms grows at a rate proportional to its current size, satisfying dy/dt = ky. If the initial population is y(0) = 500 and y(2) = 1,500, what is the population at time t = 6?

A 4,500
B 9,000
C 13,500
D 40,500

ITEM 62

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The mass of a radioactive substance decays according to dM/dt = −k M. If the half-life of the substance is 10 days and the initial mass is 80 grams, how many grams remain after 30 days?

A 20 grams
B 10 grams
C 5 grams
D 2.5 grams

ITEM 63

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A bacteria culture grows according to the differential equation dP/dt = kP, where t is measured in hours. If the population doubles every 4 hours, what is the exact value of the growth constant k?

A k = 4 ln 2
B k = (ln 4) / 2
C k = 1/4
D k = (ln 2) / 4

ITEM 64

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Newton's Law of Cooling states that dT/dt = −k(T − T_m), where T_m is the ambient temperature. A cup of coffee cools in a room at 70°F. The initial temperature is 190°F, and after 10 minutes it cools to 150°F. Which expression gives the temperature T(t) in °F for t ≥ 0?

A T(t) = 70 + 120 (2/3)^(t/10)
B T(t) = 70 + 190 (2/3)^(t/10)
C T(t) = 190 − 120 (2/3)^(t/10)
D T(t) = 70 + 120 (3/4)^(t/10)

ITEM 65

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A quantity y decays exponentially according to dy/dt = −0.05 y. If y(0) = 200, at what exact time t does y reach 50?

A t = 20 ln 2
B t = 40 ln 2
C t = 10 ln 4
D t = 50 ln 2

ITEM 66

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

An investment account balance A(t) grows continuously according to dA/dt = r A. If the account balance triples in exactly 20 years, what is the annual interest rate r?

A r = (ln 3) / 20
B r = (ln 20) / 3
C r = 20 ln 3
D r = 3 / 20

ITEM 67

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A chemical substance decomposes at a rate proportional to the amount present: dM/dt = −k M. If 20% of the substance decomposes during the first 5 hours, what fraction of the original amount remains after 10 hours?

A 60%
B 40%
C 64%
D 72%

ITEM 68

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The temperature T (in °C) of an object placed in a room at 20°C satisfies dT/dt = −0.1 (T − 20). If the initial temperature is T(0) = 100°C, what is the temperature after 10 minutes?

A 20 + 80 / e
B 20 + 100 / e
C 80 / e
D 20 + 80 e

ITEM 69

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A quantity y satisfies dy/dt = ky. If y(1) = 6 and y(3) = 24, what is the initial value y(0)?

A 1.5
B 3
C 4
D 2

ITEM 70

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

If a population grows exponentially according to dP/dt = kP and quadruples in size every 8 years, what is the doubling time of the population?

A 2 years
B 16 years
C 6 years
D 4 years

ITEM 71

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A population P(t) changes according to the logistic differential equation dP/dt = 0.04 P (1 – P / 800). What is the carrying capacity of the population, and at what population size is the population growing most rapidly?

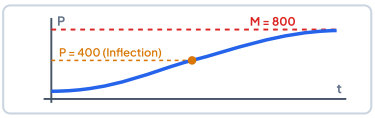


FIGURE 7.9: LOGISTIC GROWTH S-CURVE (CAPACITY M = 800)

A Carrying capacity = 800; growing most rapidly when P = 800
B Carrying capacity = 400; growing most rapidly when P = 200
C Carrying capacity = 800; growing most rapidly when P = 400
D Carrying capacity = 1,600; growing most rapidly when P = 800

ITEM 72

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A population modeled by a logistic differential equation satisfies $dP/dt = 3P - P^2 / 500$. What is the carrying capacity of this population?

A 500
B 750
C 1,500
D 3,000

ITEM 73

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The number of individuals P in a protected reserve satisfies the differential equation $dP/dt = 0.002\,P\,(600 - P)$, where t is measured in years. What is the maximum rate of population growth in individuals per year?

A 180

B 300

C 360

D 90

ITEM 74

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A population P(t) satisfies the logistic differential equation $dP/dt = 0.5\,P\,(1 - P/200)$. If $P(0) = 50$, which of the following statements about the graph of P(t) is correct?

A The graph is concave down for all $t \geq 0$.

B The graph is concave up for all $t \geq 0$.

C The graph is concave up initially, has a point of inflection when $P = 100$, and is concave down for $P > 100$.

D The graph is concave down initially, has a point of inflection when $P = 100$, and is concave up for $P > 100$.

ITEM 75

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A population satisfies $dP/dt = 0.05\,P\,(1 - P/1{,}200)$. If $P(0) = 1{,}500$, what is $\lim_{t\to\infty} P(t)$, and what is the concavity of the solution curve for $t > 0$?

A $\lim_{t\to\infty} P(t) = 1{,}200$; concave down

B $\lim_{t\to\infty} P(t) = 1{,}200$; concave up

C $\lim_{t\to\infty} P(t) = 1{,}500$; concave down

D $\lim_{t\to\infty} P(t) = 0$; concave up

ITEM 76

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A population grows logistically according to $dP/dt = kP\,(10{,}000 - P)$, with $P(0) = 1{,}000$. For what value of P is the population growing at its greatest rate?

A $P = 1{,}000$

B $P = 5{,}000$

C $P = 10{,}000$

D $P = 2{,}500$

ITEM 77

HARD · SINGLE BEST CHOICE ★★★★ MANDATORY

If $dy/dt = 2y(1 - y/50)$, which of the following is an expression for d^2y/dt^2 in terms of y?

A $4y(1 - y/25)(1 - y/50)$

B $2(1 - y/25)(1 - y/50)$

C $4y(1 - y/50)^2$

D $2(1 - y/50) - 2y/50$

ITEM 78

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A population obeys the logistic model $dP/dt = kP(1 - P/M)$. If the carrying capacity is $M = 2{,}400$ and the maximum population growth rate is 120 individuals per year, what is the value of k?

A 0.05

B 0.10

C 0.20

D 0.40

ITEM 79

HARD · SINGLE BEST CHOICE ★★★★ MANDATORY

The logistic differential equation $dP/dt = 0.1P(1 - P/500)$ models a fish population with $P(0) = 100$. What is the explicit particular solution $P(t)$?

A $P(t) = 500/(1 + 4e^{-0.1t})$

B $P(t) = 500/(1 + 5e^{-0.1t})$

C $P(t) = 100/(1 + 4e^{-0.1t})$

D $P(t) = 500/(1 + 4e^{0.1t})$

ITEM 80

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A wildlife population satisfies $dP/dt = 0.08P(1 - P/250)$. Which of the following statements is FALSE?

A If $P(0) = 200$, the graph of $P(t)$ is concave down for all $t \geq 0$.

B If $P(0) = 50$, the graph of $P(t)$ has an inflection point when $P = 125$.

C For any initial population $P(0) > 0$, $\lim_{t \to \infty} P(t) = 250$.

D The population grows fastest when $P = 250$.

ITEM 81

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $y = f(x)$ be the particular solution to the differential equation $dy/dx = xy + 1$ with the initial condition $f(1) = 2$. What is the value of d^2y/dx^2 at the point $(1, 2)$?

A 3
B 5
C 7
D 2

ITEM 82

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $y = f(x)$ be the solution to $dy/dx = 2x - y$ with $f(1) = 3$. Using the tangent line at $x = 1$, we approximate $f(1.2)$. Is this approximation an overestimate or an underestimate, and what is its value?

A 2.8; overestimate because $f''(1) < 0$
B 3.2; underestimate because $f''(1) > 0$
C 2.8; underestimate because $f''(1) > 0$
D 3.2; overestimate because $f''(1) < 0$

ITEM 83

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A tank contains 100 liters of water with 20 kg of dissolved salt. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 liters per minute. The well-stirred solution drains out at the same rate of 4 liters per minute. Which differential equation models the amount of salt $S(t)$ (in kg) in the tank at time t?



FIGURE 7.1: CONTINUOUS MIXING TANK MODEL

A $dS/dt = 2 - 0.04\,S$
B $dS/dt = 0.5 - 0.04\,S$
C $dS/dt = 2 - 4\,S$
D $dS/dt = 2 - 0.02\,S$

ITEM 84

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A tank initially holds 500 gallons of pure water. A brine solution containing 2 pounds of salt per gallon flows in at 5 gallons per minute. The well-mixed solution flows out at 3 gallons per minute. Which differential equation correctly models the amount of salt $S(t)$ (in pounds) in the tank for $t \geq 0$?

A $dS/dt = 10 - 3S / 500$
B $dS/dt = 10 - 3S / (500 + 2t)$
C $dS/dt = 10 - 5S / (500 + 2t)$
D $dS/dt = 2 - 3S / (500 + 2t)$

ITEM 85

HARD · SINGLE BEST CHOICE ★★★★ MANDATORY

If $dy/dx = y^2 / x$ for $x \neq 0$, what is d^2y/dx^2 in terms of x and y?

A $d^2y/dx^2 = (2y^3 - y^2) / x^2$

B $d^2y/dx^2 = 2y / x^2$

C $d^2y/dx^2 = (2y^3 + y^2) / x^2$

D $d^2y/dx^2 = 2y^3 / x^2$

ITEM 86

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $y = f(x)$ be the particular solution to $dy/dx = x - 2y$ with $f(0) = 1$. The tangent line at $x = 0$ is used to estimate $f(0.1)$. What is the estimate, and is it an underestimate or an overestimate?

A 0.8; underestimate because $f''(0) = 5 > 0$

B 0.8; overestimate because $f''(0) = -5 < 0$

C 1.2; underestimate because $f''(0) = 5 > 0$

D 1.2; overestimate because $f''(0) = -5 < 0$

ITEM 87

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

A tank contains 200 liters of fluid. A solution containing 3 grams per liter of dye flows in at 4 liters per minute, and the well-stirred mixture flows out at 4 liters per minute. What is the limiting amount of dye in the tank as $t \to \infty$?

A 200 grams

B 600 grams

C 800 grams

D 1,200 grams

ITEM 88

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $y = f(x)$ be a solution curve to $dy/dx = (x + 1)(y - 2)$. For which points in the xy-plane does the solution curve have a relative minimum?

A All points on the line $y = 2$

B Points of the form $(-1, y)$ where $y > 2$

C Points of the form $(-1, y)$ where $y < 2$

D Points of the form $(x, 2)$ where $x > -1$

ITEM 89

HARD · SINGLE BEST CHOICE ★★★★ MANDATORY

If $dy/dx = e^y \sin x$, which of the following is equal to d^2y/dx^2?

A $e^{2y} \sin^2 x + e^y \cos x$

B $e^y \sin^2 x + e^y \cos x$

C $e^{2y} \cos x$

D $e^y (\cos x + \sin x)$

ITEM 90

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $x > 0$, what is the geometric nature of the family of general solutions to the differential equation $dy/dx = (y - 3) / x$?

A A family of parabolas with vertices at (0, 3)

B A family of exponential curves asymptotic to $y = 3$

C A family of non-vertical lines passing through the point (0, 3)

D A family of concentric circles centered at (0, 3)

FRQ 1: Separation of Variables, Tangent Lines & Concavity

★★★★★ VERY IMPORTANT | 9 Points

Consider the differential equation $dy/dx = (x - 1)y^2$ with a particular solution $y = f(x)$ satisfying the initial condition $f(2) = 1$.

(a) [2 Points] Write an equation for the line tangent to the graph of f at $x = 2$, and use it to approximate $f(2.1)$.

(b) [2 Points] Find d^2y/dx^2 at the point $(2, 1)$. Determine whether the approximation calculated in part (a) is an overestimate or an underestimate of $f(2.1)$. Justify your answer.

(c) [5 Points] Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(2) = 1$, and state its domain.

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

FRQ 2: Euler's Method & Solution Analysis (BC Exclusive)

★★★★★ VERY IMPORTANT | 9 Points

Let $y = f(x)$ be the particular solution to the differential equation $dy/dx = 2x - y$ with initial condition $f(1) = 2$.

Step	Current (x, y)	Slope $dy/dx = 2x - y$	Step Size Δx	Increment Δy	Next (x, y)
1	(1.0, 2.0)	$2(1) - 2 = 0$	0.5	$0(0.5) = 0$	(1.5, 2.0)
2	(1.5, 2.0)	$2(1.5) - 2 = 1.0$	0.5	$1.0(0.5) = 0.5$	**(2.0, 2.5)**

(a) [3 Points] Use Euler's method with two steps of equal size $\Delta x = 0.5$, starting at $x = 1$, to approximate $f(2)$. Show the computations that lead to your answer.

(b) [2 Points] Find d^2y/dx^2 in terms of x and y. Determine whether the approximation for $f(2)$ obtained in part (a) is an overestimate or an underestimate. Justify your answer.

(c) [2 Points] Find the constants m and b such that $y = mx + b$ is an exact solution to the differential equation $dy/dx = 2x - y$.

(d) [2 Points] Let $g(x) = f(x) - (2x - 2)$. Given that $\lim_{x \to \infty} g(x) = 0$, evaluate $\lim_{x \to \infty} [f(x) / x]$.

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

FRQ 3: Logistic Population Dynamics (BC Exclusive)

★★★★★ VERY IMPORTANT 9 Points

A population of protected wild elk is introduced into a forest reserve. The population P(t) satisfies the differential equation $\frac{dP}{dt} = \frac{1}{200} P(1{,}000 - P)$ for $t \geq 0$, where t is measured in years and $P(0) = 200$.

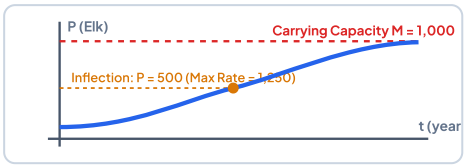


FIGURE FRQ-3: LOGISTIC POPULATION DYNAMICS $dP/dt = (1/200)\,P\,(1{,}000 - P)$

(a) [2 Points] For what values of P is the population increasing? For what values of P is the population decreasing? Justify your answers.

(b) [2 Points] At what population size P is the population growing most rapidly? Find this maximum rate of growth in elk per year.

(c) [2 Points] Evaluate $\lim_{t \to \infty} P(t)$ and $\lim_{t \to \infty} \frac{dP}{dt}$. Explain the biological meaning of both limits.

(d) [3 Points] Use separation of variables to find the explicit particular solution P(t) for $t \geq 0$.

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

FRQ 4: Slope Fields & Particular Solutions with Radical Branches

★★★★★ VERY IMPORTANT

9 Points

Consider the differential equation $dy/dx = (x + 2) / y$ for $y \neq 0$, with particular solution $y = f(x)$ satisfying $f(0) = -2$.

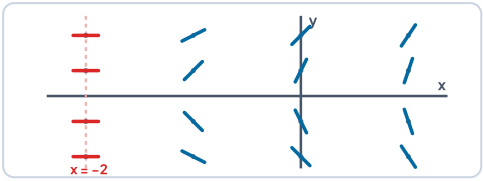


FIGURE FRQ-4: OFFICIAL AP SLOPE FIELD GRID FOR DY/DX = (X + 2) / Y (12 SELECTED POINTS)

(a) [2 Points] Describe the behavior of the slopes of the solution curves along the line $x = -2$ and along the line $y = 0$.

(b) [2 Points] Write an equation for the line tangent to the graph of f at $x = 0$, and use it to approximate $f(0.2)$.

(c) [5 Points] Find the particular solution $y = f(x)$ to the differential equation with $f(0) = -2$, and state its largest open domain of validity.

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

FRQ 5: Mixing Tank Differential Equation & Steady-State

★★★★ MANDATORY 9 Points

A large mixing tank contains 100 gallons of solution with 50 pounds of dissolved salt at time t = 0. Brine containing 2 pounds of salt per gallon enters the tank at a constant rate of 3 gallons per minute. The well-mixed solution drains out of the tank at the same rate of 3 gallons per minute.

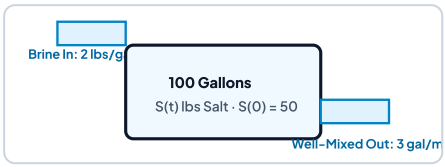


FIGURE FRQ-5: SCHEMATIC DIAGRAM FOR 100-GALLON MIXING TANK PROBLEM

(a) [2 Points] Write a differential equation that models the rate of change of salt S(t) (in pounds) in the tank with respect to time t (in minutes). State the initial condition.

(b) [4 Points] Find the particular solution S(t) by separation of variables.

(c) [1 Points] Determine the exact amount of salt in the tank at time t = 10 minutes.

(d) [2 Points] Evaluate $\lim_{t\to\infty} S(t)$, and provide a physical explanation for this limiting value based on the problem context.

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

FRQ 6: Euler's Method & Taylor Polynomial Comparison (BC Exclusive)

★★★★★ VERY IMPORTANT

9 Points

Consider the differential equation $dy/dx = x^2 + y$ with the initial condition $f(0) = 1$.

(a) [3 Points] Use Euler's method with two equal steps of $\Delta x = 0.2$, starting at $x = 0$, to approximate $f(0.4)$. Show all steps.

(b) [2 Points] Find d^2y/dx^2 in terms of x and y. Evaluate d^2y/dx^2 at $(0, 1)$.

(c) [2 Points] Is the approximation for $f(0.4)$ obtained in part (a) greater than or less than the actual value of $f(0.4)$? Justify your answer.

(d) [2 Points] Write the second-degree Taylor polynomial for f centered at $x = 0$, and use it to compute an approximation for $f(0.4)$.

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

FRQ 7: Newton's Law of Cooling & Rate Analysis

★★★★ MANDATORY 9 Points

A heated metal bar is placed into an environment with a constant ambient temperature of 65°F. The temperature T(t) of the bar (in °F) cools according to Newton's Law of Cooling: $dT/dt = -k(T - 65)$, where t is measured in minutes. At time $t = 0$, the bar has a temperature of 185°F, and at time $t = 10$, its temperature is 125°F.

(a) [3 Points] Use separation of variables to show that $T(t) = 65 + 120\,e^{-kt}$.

(b) [2 Points] Find the exact value of the cooling constant k.

(c) [2 Points] At what time t will the temperature of the metal bar reach 85°F? Express your answer in terms of natural logarithms.

(d) [2 Points] Find the value of dT/dt at time $t = 10$ minutes, and interpret its meaning including units.

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

FRQ 8: Separation of Variables with Inverse Trigonometric Forms

★★★★★ VERY IMPORTANT | 9 Points

Consider the differential equation $dy/dx = (1 + y^2)\,e^x$ with initial condition $y(0) = 1$.

(a) [1 Points] Find the slope of the line tangent to the graph of y at the point where $x = 0$.

(b) [2 Points] Find d^2y/dx^2 at the point where $x = 0$. Determine whether the graph of the solution is concave up or concave down at this point.

(c) [4 Points] Find the particular solution $y = f(x)$ satisfying the initial condition $f(0) = 1$.

(d) [2 Points] Determine the domain of the particular solution $y = f(x)$.

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

FRQ 9: Comparative Population Growth Models (BC Exclusive)

★★★★★ VERY IMPORTANT | 9 Points

Two biological models are proposed for the growth of a colony of micro-organisms, both with the initial condition of 50 organisms at t = 0 (t in days):
• Model A (Exponential): $dy/dt = 0.2y$
• Model B (Logistic): $dP/dt = 0.2P(1 - P/500)$

(a) [2 Points] Find the explicit solution y(t) for Model A, and find the doubling time of the population under Model A.

(b) [3 Points] For Model B, determine the carrying capacity of the colony, the population size at which the colony grows most rapidly, and the maximum growth rate in organisms per day.

(c) [2 Points] Evaluate $\lim_{t\to\infty} [(dP/dt)/(dy/dt)]$. Justify your answer.

(d) [2 Points] Describe the behavior of y(t) compared to P(t) as t becomes arbitrarily large. Which model is more realistic for an environment with limited resources, and why?

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

FRQ 10: Differential Equations Synthesis Masterpiece

★★★★★ VERY IMPORTANT 9 Points

Consider the differential equation $dy/dx = (x\,y) / 2$ for $y > 0$ with particular solution $y = f(x)$ satisfying $f(0) = 3$.

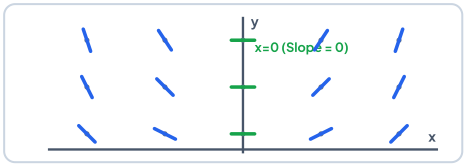


FIGURE FRQ-10: SLOPE FIELD GRID FOR DY/DX = (XY) / 2 (UPPER HALF-PLANE Y > 0)

(a) [1 Points] Find all points (x, y) where the slope of the solution curve is equal to 0.

(b) [2 Points] Use Euler's method with two steps of equal size $\Delta x = 0.5$, starting at $x = 0$, to approximate $f(1)$.

(c) [2 Points] Find d^2y/dx^2 in terms of x and y. Determine whether the graph of f has a relative minimum, relative maximum, or neither at $x = 0$. Justify your answer.

(d) [4 Points] Find the exact particular solution $y = f(x)$ with $f(0) = 3$. Compute $f(1)$ and compare it to the Euler approximation from part (b).

Official AP Scoring Directive: Show all supporting work. Justify conclusions with specific mathematical statements and named calculus theorems.

End of Question Paper

You have concluded Questions 1 through 100 of the Unit 7 Master Examination Repository.

Important Study Strategy: Carefully compare your solutions against the detailed answer keys and explanations that follow. Pay special attention to why incorrect choices were selected as distractors and how full credit is awarded on official AP rubrics.

Answer Section Contents

- **Answer Key Table:** Complete quick-reference grid of letter choices for Questions 1–90.
- **MCQ Detailed Solutions:** Step-by-step bulleted solutions for Questions 1–90 with full distractor error analysis.
- **FRQ Model Solutions:** Full analytical solutions for Questions 91–100 with official AP 9-point rubric point breakdowns.
- **Formula Appendix:** Summary of high-yield Unit 7 differential equations theorems and formulas.

Multiple Choice Answer Key

Verified answer keys with priority ratings. Questions 91–100 are free response (90 points total).

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
B ★★★	C ★★★	A ★★★	B ★★★	C ★★★	C ★★★★★	A ★★★	B ★★★	C ★★★★★	A ★★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
B ★★★	D ★★★	A ★★★★★	C ★★★★	B ★★★★★	A ★★★★★	D ★★★★	C ★★★★★	A ★★★★★	B ★★★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
C ★★★★★	B ★★★★★	D ★★★	A ★★★★★	C ★★★★★	D ★★★★★	B ★★★★	A ★★★★	C ★★★	D ★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
C ★★★★★	A ★★★★	D ★★★★★	B ★★★★★	A ★★★★	C ★★★★	D ★★★★★	B ★★★★★	A ★★★★★	D ★★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
B ★★★★★	D ★★★★★	C ★★★★★	A ★★★★	B ★★★★★	C ★★★★★	B ★★★★	D ★★★★★	A ★★★★	C ★★★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
B ★★★★★	A ★★★★★	C ★★★★	D ★★★★★	C ★★★★★	D ★★★★	C ★★★★★	B ★★★★	B ★★★★★	D ★★★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
C ★★★★★	B ★★★★★	D ★★★★	A ★★★★★	B ★★★★	A ★★★★★	C ★★★★	A ★★★★★	B ★★★★★	D ★★★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
C ★★★★★	C ★★★★★	A ★★★★★	C ★★★★	B ★★★★★	B ★★★★★	A ★★★★	C ★★★★★	A ★★★★	D ★★★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
B ★★★★★	C ★★★★★	A ★★★★	B ★★★★★	A ★★★★	A ★★★★★	B ★★★★	B ★★★★★	A ★★★★	C ★★★★★

Scoring Summary: Each multiple choice question is worth 1 point (90 points total). Each free response question is worth 9 points (90 points total). Total Exam Value: 180 Points.

Question 01 Analysis

★★★ IMPORTANT

Correct Answer: (B) $dV/dt = -k\,S$

STEP-BY-STEP SOLUTION

- **Identify Rate of Change:** The rate of change of volume with respect to time is represented by the derivative dV/dt.
- **Direct Proportionality:** Directly proportional to surface area S means $dV/dt = C \cdot S$ for some proportionality constant C.
- **Physical Melting Condition:** Because the snowball is melting, its volume is decreasing over time ($dV/dt < 0$). For a positive constant k, the constant of proportionality must be negative: $C = -k$.
- **Conclusion:** The correct differential equation is $dV/dt = -k\,S$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Models an expanding snowball where volume increases with surface area (omits the negative sign for melting).

✗ **(C)** Models inverse proportionality rather than direct proportionality.

✗ **(D)** Inverts the roles of volume V and surface area S.

Question 02 Analysis

★★★ IMPORTANT

Correct Answer: (C) $dP/dt = k\sqrt{P}$

STEP-BY-STEP SOLUTION

- **Rate of Growth:** The growth rate of the bacteria population is given by dP/dt.
- **Proportionality Translation:** Directly proportional to the square root of P means $dP/dt = k \cdot \sqrt{P}$.
- **Conclusion:** With positive constant k, the model is $dP/dt = k\sqrt{P}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the square of the population (P^2) instead of the square root ($\sqrt{P}$).

✗ **(B)** Represents inverse proportionality to $\sqrt{P}$.

✗ **(D)** Takes the square root of the constant k rather than the population P.

Question 03 Analysis

★★★ MANDATORY

Correct Answer: (A) $dT/dt = -k\,(T - M)$

STEP-BY-STEP SOLUTION

- **Temperature Difference:** The difference between the object temperature T and the surrounding room temperature M is $(T - M)$.
- **Direction of Heat Transfer:** Since $T > M$, the object cools down, meaning its temperature decreases: $dT/dt < 0$.
- **Negative Constant:** For $k > 0$ and $(T - M) > 0$, the rate must be negative: $dT/dt = -k\,(T - M)$.
- **Conclusion:** The correct cooling equation is $dT/dt = -k\,(T - M)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds the temperatures $(T + M)$ instead of taking their difference.

✗ **(C)** When $T > M$, $(M - T)$ is already negative; with $-k$, this would falsely predict that the object heats up.

✗ **(D)** Multiplies the temperatures together instead of taking their difference.

Question 04 Analysis

★★★ MANDATORY

Correct Answer: (B) $y = e^{3x}$

STEP-BY-STEP SOLUTION

- **Differentiating $y = e^{3x}$:** Using the chain rule, $dy/dx = d/dx\,[e^{3x}] = 3e^{3x}$.
- **Comparing to 3y:** Since $y = e^{3x}$, we have $3y = 3e^{3x}$.
- **Verification:** Because $dy/dx = 3y$ holds identically for all x, $y = e^{3x}$ is a solution.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** For $y = 3e^x$, $dy/dx = 3e^x$, but $3y = 9e^x$; $dy/dx \neq 3y$.

✗ **(C)** For $y = x^3$, $dy/dx = 3x^2$, while $3y = 3x^3$; $dy/dx \neq 3y$.

✗ **(D)** For $y = \sin(3x)$, $dy/dx = 3\cos(3x)$, while $3y = 3\sin(3x)$; $dy/dx \neq 3y$.

Question 05 Analysis

★★★ MANDATORY

Correct Answer: (C) $y = \cos(2x)$

STEP-BY-STEP SOLUTION

- **First Derivative:** For $y = \cos(2x)$, $dy/dx = -2\sin(2x)$.
- **Second Derivative:** $y'' = d/dx\,[-2\sin(2x)] = -4\cos(2x)$.
- **Substitution:** $y'' + 4y = -4\cos(2x) + 4(\cos(2x)) = 0$.
- **Conclusion:** $y = \cos(2x)$ satisfies the differential equation.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** For $y = e^{2x}$, $y'' = 4e^{2x}$, so $y'' + 4y = 8e^{2x} \neq 0$.

✗ **(B)** For $y = e^{-2x}$, $y'' = 4e^{-2x}$, so $y'' + 4y = 8e^{-2x} \neq 0$.

✗ **(D)** For $y = \cos(4x)$, $y'' = -16\cos(4x)$, so $y'' + 4y = -12\cos(4x) \neq 0$.

Question 06 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) Any real number k

STEP-BY-STEP SOLUTION

- **Compute Derivative:** For $y = x^2 + k\,x^{-1}$, $dy/dx = 2x - k\,x^{-2} = 2x - k/x^2$.
- **Substitute into Left Side:** $x \cdot (dy/dx) + y = x(2x - k/x^2) + (x^2 + k/x)$.
- **Algebraic Simplification:** $(2x^2 - k/x) + (x^2 + k/x) = 3x^2$.
- **Conclusion:** The terms involving k cancel out completely ($-k/x + k/x = 0$). The equation holds for ANY real number k.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Incorrectly assumes the singular term k/x must be zero to satisfy the equation.

✗ **(B)** Confuses the coefficient 3 in $3x^2$ with the value of k.

✗ **(D)** Fails to recognize that $-k/x$ and $+k/x$ cancel identically.

Question 07 Analysis

★★★ MANDATORY

Correct Answer: (A) $dy/dx = -2y + 6$

STEP-BY-STEP SOLUTION

- **Compute Derivative:** $dy/dx = d/dx\,[C e^{-2x} + 3] = -2C e^{-2x}$.
- **Express in Terms of y:** From the given equation, $C e^{-2x} = y - 3$.
- **Substitute:** $dy/dx = -2(y - 3) = -2y + 6$.
- **Conclusion:** The differential equation is $dy/dx = -2y + 6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error when distributing -2 across $(y - 3)$: writes $-2y - 6$ instead of $-2y + 6$.
✗ **(C)** Drops the negative sign from the exponent derivative: uses $+2$ instead of -2.
✗ **(D)** Forgets to multiply the constant 3 by the factor -2.

Question 08 Analysis

★★★ IMPORTANT

Correct Answer: (B) $dh/dt = -k / h^2$

STEP-BY-STEP SOLUTION

- **Inverse Proportionality:** Inversely proportional to h^2 means the rate has the form C / h^2.
- **Draining Condition:** Because the tank is draining, the water depth is decreasing over time ($dh/dt < 0$).
- **Applying Constant:** For positive constant k, the proportionality coefficient must be negative: $dh/dt = -k / h^2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Represents direct proportionality to h^2 rather than inverse proportionality.
✗ **(C)** Omits the negative sign, which would model water filling the tank rather than draining.
✗ **(D)** Uses the square root of depth ($\sqrt{h}$) instead of the square of depth (h^2).

Question 09 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $y = 5e^{x^2}$

STEP-BY-STEP SOLUTION

- **Test Differential Equation:** For $y = 5e^{x^2}$, $dy/dx = 5 \cdot d/dx[e^{x^2}] = 5 \cdot (2x\, e^{x^2}) = 2x(5e^{x^2}) = 2x\, y$. This satisfies $dy/dx = 2x\, y$.
- **Test Initial Condition:** $y(0) = 5e^{0^2} = 5(1) = 5$.
- **Conclusion:** $y = 5e^{x^2}$ satisfies both the differential equation and the initial condition.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** For $y = 5e^{2x}$, $dy/dx = 10e^{2x} = 2y$, which does not contain the variable factor x.
✗ **(B)** Satisfies $y(0) = 1 + 4 = 5$, but $dy/dx = 2x\, e^{x^2} \neq 2x\, y$ because of the constant $+4$.
✗ **(D)** For $y = 5x^2 + 5$, $dy/dx = 10x$, but $2x\, y = 2x(5x^2 + 5) = 10x^3 + 10x \neq 10x$.

Question 10 Analysis

★★★★ MANDATORY

Correct Answer: (A) $x \cdot (dy/dx) - y = x$

STEP-BY-STEP SOLUTION

- **Compute Derivative:** Use the Product Rule on $y = x \ln x$: $dy/dx = (1)\cdot\ln x + x\cdot(1/x) = \ln x + 1$.
- **Substitute into $x \cdot (dy/dx) - y$:** $x(\ln x + 1) - x \ln x = x \ln x + x - x \ln x = x$.
- **Verification:** The expression simplifies identically to x, so $x \cdot (dy/dx) - y = x$ is satisfied.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Computes $x(\ln x + 1) + x \ln x = 2x \ln x + x \neq x$.
✗ **(C)** Omits the product rule term $+1$ from differentiating x.
✗ **(D)** Computes $x^2(\ln x + 1) \neq x \ln x$.

Question 11 Analysis

★★★ MANDATORY

Correct Answer: (B) $dy/dx = y - 1$

STEP-BY-STEP SOLUTION

- **Geometric Condition:** Having the same slope along any horizontal line $y = c$ means that dy/dx does not change as x varies.
- **Variable Dependence:** The slope dy/dx must be independent of x and depend solely on y.
- **Evaluating Candidates:** In $dy/dx = y - 1$, the slope depends purely on y, so for any fixed $y = c$, the slope is constant.
- **Conclusion:** The correct differential equation is $dy/dx = y - 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** $dy/dx = 2x$ produces identical slopes along vertical lines (constant x), not horizontal lines.
✗ **(C)** $dy/dx = x + y$ depends on both variables, so slopes change along both horizontal and vertical lines.
✗ **(D)** $dy/dx = xy$ changes as x varies along any horizontal line $y = c$ ($c \neq 0$).

Question 12 Analysis

★★★ MANDATORY

Correct Answer: (D) $dy/dx = x - 2$

STEP-BY-STEP SOLUTION

- **Invariance Condition:** Slopes are identical along any vertical line $x = k$. This requires that dy/dx is independent of y.
- **Variable Analysis:** The formula for dy/dx must contain only the variable x (or be a constant).
- **Examining Choices:** For $dy/dx = x - 2$, the value of the derivative is determined strictly by x, remaining completely constant as y changes.
- **Conclusion:** $dy/dx = x - 2$ produces the described slope field.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** $dy/dx = y^2$ depends entirely on y, meaning slopes are constant along horizontal lines.
✗ **(B)** $dy/dx = x / y$ depends on both variables and is undefined along the x-axis ($y = 0$).
✗ **(C)** $dy/dx = x + y$ depends on both variables, so slopes vary along vertical lines.

Question 13 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $y = x / 2$

STEP-BY-STEP SOLUTION

- **Horizontal Tangent Condition:** A line segment is horizontal if and only if its slope equals zero: $dy/dx = 0$.
- **Setting the Derivative to Zero:** $x - 2y = 0$.
- **Solving for y:** $2y = x \Rightarrow y = x / 2$.
- **Conclusion:** All tangent segments are horizontal along the line $y = x / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Inverts the coefficient: solves $2x - y = 0$ giving $y = 2x$.
✗ **(C)** Sign error: solves $x + 2y = 0$ giving $y = -x / 2$.
✗ **(D)** $x = 0$ is the y-axis, along which $dy/dx = -2y$, which is not zero everywhere.

Question 14 Analysis

★★★★ MANDATORY

Correct Answer: (C) Quadrants III and IV only

STEP-BY-STEP SOLUTION

- **Sign of x^2:** For all points not on the y-axis ($x \neq 0$), $x^2 > 0$.
- **Condition for Negative Slope:** $dy/dx = x^2 y < 0$ requires $y < 0$, since x^2 is strictly positive.
- **Identifying Quadrants:** The region where $y < 0$ is below the x-axis, which corresponds to Quadrant III ($x < 0, y < 0$) and Quadrant IV ($x > 0, y < 0$).
- **Conclusion:** Slopes are strictly negative in Quadrants III and IV.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** In Quadrant II, $x < 0$ and $y > 0$, so $dy/dx = (+)(+) =$ positive.
✗ **(B)** Omits Quadrant III, where $x < 0$ but x^2 is still positive and $y < 0$.
✗ **(D)** Includes Quadrant II, where slopes are positive.

Question 15 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) –1

STEP-BY-STEP SOLUTION

- **First Derivative at Initial Point:** At (1, 1), $dy/dx = 1 - 1 = 0$.
- **Second Derivative Expression:** Differentiate $dy/dx = y - x$ implicitly with respect to x: $d^2y/dx^2 = dy/dx - 1$.
- **Substitution:** At (1, 1), $f''(1) = (dy/dx)|_{(1,1)} - 1 = 0 - 1 = -1$.
- **Conclusion:** $f''(1) = -1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses $f''(1)$ with $f'(1) = 0$.
✗ **(C)** Sign error in differentiating $-x$: treats $d/dx[-x]$ as $+1$.
✗ **(D)** Evaluates $1 - (-1) = 2$.

Question 16 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) Concave up, because f''(1) = 4 > 0

STEP-BY-STEP SOLUTION

- **First Derivative at (1, 2):** $f'(1) = (1)(2) = 2$.
- **Implicit Second Derivative:** Apply the Product Rule: $d^2y/dx^2 = d/dx\,[x\,y] = (1)(y) + x(dy/dx) = y + x(dy/dx)$.
- **Evaluate at (1, 2):** $f''(1) = 2 + (1)(2) = 4$.
- **Concavity Determination:** Since $f''(1) = 4 > 0$, the graph of f is strictly concave up at $x = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error resulting in −4.
✗ **(C)** Assumes the derivative is constant.
✗ **(D)** The second derivative can always be computed directly using implicit differentiation without an explicit closed form for f(x).

Question 17 Analysis

★★★★ MANDATORY

Correct Answer: (D) $y = 6x + 3$

STEP-BY-STEP SOLUTION

- **Point of Tangency:** Given $f(0) = 3$, the point is $(0, 3)$.
- **Slope from Differential Equation:** $dy/dx|_{(0,\,3)} = 0 + 2(3) = 6$.
- **Point-Slope Form:** $y - 3 = 6(x - 0) \Rightarrow y = 6x + 3$.
- **Conclusion:** The tangent line equation is $y = 6x + 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Swaps the slope 6 and y-intercept 3: $y = 3x + 6$.
✗ **(B)** Uses only the coefficient of y as the slope: $m = 2$.
✗ **(C)** Uses $m = 1$ from the x-coefficient.

Question 18 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $dy/dx = 2x^2 + y^2 - 4$

STEP-BY-STEP SOLUTION

- **Horizontal Tangent Definition:** All tangent segments must satisfy $dy/dx = 0$.
- **Condition on the Ellipse:** For all points (x, y) lying on $2x^2 + y^2 = 4$, we have $2x^2 + y^2 - 4 = 0$.
- **Matching Differential Equation:** If $dy/dx = 2x^2 + y^2 - 4$, then whenever (x, y) is on the ellipse, $dy/dx = 0$.
- **Conclusion:** $dy/dx = 2x^2 + y^2 - 4$ has horizontal tangents exactly on the ellipse.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** On the ellipse, $2x^2 + y^2 = 4$, so dy/dx would equal 4, not 0.
✗ **(B)** $dy/dx = 4 - 2x^2$ equals 0 only when $x = \pm\sqrt{2}$, which gives vertical lines, not the full ellipse.
✗ **(D)** $y^2 - 2x^2 = 0$ defines a pair of intersecting lines ($y = \pm x\sqrt{2}$), not an ellipse.

Question 19 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) f(x) = 4

STEP-BY-STEP SOLUTION

- **Equilibrium (Constant) Solution:** Notice that $y = 4$ makes the factor $(y - 4)$ equal to zero for all x.
- **Derivative of Constant Function:** If $y = 4$, then $dy/dx = 0$. Substituting $y = 4$ into the right side gives $(4 - 4)(x^2 + 1) = 0$ for all x.
- **Initial Condition Check:** The constant solution $y = 4$ satisfies $y(0) = 4$.
- **Uniqueness of Solutions:** By the Existence and Uniqueness Theorem, the unique solution through $(0, 4)$ is the constant function $f(x) = 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Introduces an exponential term when the rate of change is identically zero.

✗ **(C)** For $y = x^2 + 4$, $dy/dx = 2x$, while $(y - 4)(x^2 + 1) = x^2(x^2 + 1) \neq 2x$.

✗ **(D)** Attempts an unnecessary separation of variables without checking the zero-root equilibrium.

Question 20 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) Relative minimum

STEP-BY-STEP SOLUTION

- **First Derivative at (3, 1):** $dy/dx|_{(3,\,1)} = (3 - 3)(1 + 2) = 0 \cdot 3 = 0$. Thus, $x = 3$ is a critical point.
- **Compute Second Derivative:** Differentiate using the Product Rule: $d^2y/dx^2 = d/dx\,[(x - 3)(y + 2)] = (1)(y + 2) + (x - 3)\cdot(dy/dx)$.
- **Evaluate at (3, 1):** $d^2y/dx^2|_{(3,\,1)} = (1)(1 + 2) + (0)(0) = 3$.
- **Second Derivative Test:** Since $dy/dx = 0$ and $d^2y/dx^2 = 3 > 0$, the curve is concave up, meaning the solution has a relative minimum at $x = 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Concludes relative maximum by misinterpreting $d^2y/dx^2 > 0$ as concave down.

✗ **(C)** A point of inflection requires $d^2y/dx^2 = 0$ or a sign change, whereas $d^2y/dx^2 = 3 \neq 0$.

✗ **(D)** The derivative is fully defined and equal to 0 at (3, 1).

Question 21 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2.50

STEP-BY-STEP SOLUTION

- **Initial Point:** Start at $(x_0, y_0) = (0, 1)$ with step size $\Delta x = 0.5$.
- **Step 1:**
  - Slope: $m_0 = dy/dx|_{(0,\,1)} = 0 + 1 = 1$.
  - New y: $y_1 = y_0 + m_0 \cdot \Delta x = 1 + 1(0.5) = 1.5$.
  - New x: $x_1 = 0 + 0.5 = 0.5$.
- **Step 2:**
  - Slope: $m_1 = dy/dx|_{(0.5,\,1.5)} = 0.5 + 1.5 = 2.0$.
  - New y: $y_2 = y_1 + m_1 \cdot \Delta x = 1.5 + 2.0(0.5) = 1.5 + 1.0 = 2.5$.
  - New x: $x_2 = 0.5 + 0.5 = 1.0$.
- **Conclusion:** The Euler approximation for $f(1)$ is 2.50.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates only a single step of width 1.0: $1 + 1(1) = 2.00$.

✗ **(B)** Uses $m_1 = 1.5$ instead of 2.0: $1.5 + 1.5(0.5) = 2.25$.

✗ **(D)** Uses slope $m_1 = 2.5$ incorrectly: $1.5 + 2.5(0.5) = 2.75$.

Question 22 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1.08

STEP-BY-STEP SOLUTION

- **Step 1 (from x = 0 to x = 0.2):**
  - $(x_0, y_0) = (0, 1)$.
  - Slope: $m_0 = 2(0)(1) = 0$.
  - $y_1 = 1 + 0(0.2) = 1$.
  - Point: $(0.2, 1)$.
- **Step 2 (from x = 0.2 to x = 0.4):**
  - Slope: $m_1 = 2(0.2)(1) = 0.4$.
  - $y_2 = 1 + 0.4(0.2) = 1 + 0.08 = 1.08$.
  - Point: $(0.4, 1.08)$.
- **Conclusion:** The approximation for $f(0.4)$ is 1.08.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Repeats the zero slope from the first step and reports 1.00.

✗ **(C)** Multiplies m_1 by 0.4 instead of the step size $\Delta x = 0.2$: $1 + 0.16 = 1.16$.

✗ **(D)** Adds the slope directly to y without multiplying by Δx: $1 + 0.2 = 1.20$.

Question 23 Analysis

★★★ MANDATORY

Correct Answer: (D) 2.2

STEP-BY-STEP SOLUTION

- **Connection to Tangent Line:** A single-step Euler approximation is identical to the local linear (tangent line) approximation at x = 1.
- **Compute Slope:** $m = dy/dx|_{(1,\,2)} = 3(1) - 2 = 1$.
- **Euler Formula:** $y(1.2) \approx y(1) + m \cdot \Delta x = 2 + 1(0.2) = 2.2$.
- **Conclusion:** $f(1.2) \approx 2.2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes no change in y over the step: reports 2.0.

✗ **(B)** Uses slope m = 2 instead of 1: $2 + 2(0.2) = 2.4$.

✗ **(C)** Uses m = 3: $2 + 3(0.2) = 2.6$.

Question 24 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 2.25

STEP-BY-STEP SOLUTION

- **Negative Step Concept:** When stepping backwards from x = 2 to x = 1, $\Delta x = (1 - 2)/2 = -0.5$.
- **Step 1 (x = 2 to x = 1.5):**
  - $(x_0, y_0) = (2, 3)$.
  - Slope: $m_0 = 2(2) - 3 = 1$.
  - $y_1 = 3 + 1(-0.5) = 2.5$.
  - Point: $(1.5, 2.5)$.
- **Step 2 (x = 1.5 to x = 1.0):**
  - Slope: $m_1 = 2(1.5) - 2.5 = 3 - 2.5 = 0.5$.
  - $y_2 = 2.5 + 0.5(-0.5) = 2.5 - 0.25 = 2.25$.
  - Point: $(1.0, 2.25)$.
- **Conclusion:** $f(1) \approx 2.25$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Stops after the first step and reports $y_1 = 2.50$.

✗ **(C)** Adds $\Delta x = 0.5$ without accounting for the negative direction: $2.5 - 0.5 = 2.00$.

✗ **(D)** Makes an arithmetic error: subtracts 0.5 twice.

Question 25 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) Underestimate, because f"(x) > 0 on [0, 0.5]

STEP-BY-STEP SOLUTION

- **Compute Second Derivative:** Differentiate implicitly: $d^2y/dx^2 = d/dx\,[x^2 + y] = 2x + dy/dx = 2x + x^2 + y$.
- **Sign of Second Derivative:** On [0, 0.5], $x \geq 0$ and $y \geq 1$ (since $dy/dx > 0$ so y increases from 1). Therefore, $d^2y/dx^2 = 2x + x^2 + y > 0$ for all x in [0, 0.5].
- **Concavity and Tangent Lines:** Since f"(x) > 0, the curve is strictly concave up. Tangent line segments lie strictly BELOW the curve of a concave up function.
- **Conclusion:** Euler's method underestimates the true value because f"(x) > 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Incorrectly claims that concave up causes an overestimate (it causes an underestimate).

✗ **(B)** Attributes the error direction to the first derivative rather than concavity (second derivative).

✗ **(D)** Confuses the role of f'(x) with the concavity test for tangent lines.

Question 26 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 8.875

STEP-BY-STEP SOLUTION

- **Step 1 (x = 1.0 to x = 1.5):**
  - $(x_0, y_0) = (1, 2)$.
  - Slope: $m_0 = 2^2 - 1 = 4 - 1 = 3$.
  - $y_1 = 2 + 3(0.5) = 3.5$.
  - Point: (1.5, 3.5).
- **Step 2 (x = 1.5 to x = 2.0):**
  - Slope: $m_1 = (3.5)^2 - 1.5 = 12.25 - 1.5 = 10.75$.
  - $y_2 = 3.5 + 10.75(0.5) = 3.5 + 5.375 = 8.875$.
  - Point: (2.0, 8.875).
- **Conclusion:** The approximation for y(2) is 8.875.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reports only the increment of the second step $\Delta y_2 = 5.375$ without adding $y_1 = 3.5$.

✗ **(B)** Uses $m_1 = 7.5$ instead of 10.75.

✗ **(C)** Reports the slope $m_1 = 10.75$ instead of the updated y-value.

Question 27 Analysis

★★★★ MANDATORY

Correct Answer: (B) 5.5

STEP-BY-STEP SOLUTION

- **Identify Given Information:** At x = 0.5, the current y-value is 3.5 and the slope is dy/dx = 4.
- **Step Size:** $\Delta x = 1.0 - 0.5 = 0.5$.
- **Euler Update Formula:** $k = y(0.5) + (dy/dx)|_{(0.5,\ 3.5)} \cdot \Delta x$.
- **Calculation:** $k = 3.5 + 4(0.5) = 3.5 + 2.0 = 5.5$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses slope from the previous step (3): $3.5 + 3(0.5) = 5.0$.

✗ **(C)** Adds the full slope 4 without multiplying by $\Delta x = 0.5$: $3.5 + 4 = 7.5$ or $2 + 4 = 6.0$.

✗ **(D)** Adds the slope 4 directly: $3.5 + 4 = 7.5$.

Question 28 Analysis

★★★★ MANDATORY

Correct Answer: (A) 0.06

STEP-BY-STEP SOLUTION

- **Step 1 (x = 0 to 0.1):** (0, 0). Slope $m_0 = 2(0) = 0$. $y_1 = 0 + 0(0.1) = 0$.
- **Step 2 (x = 0.1 to 0.2):** (0.1, 0). Slope $m_1 = 2(0.1) = 0.2$. $y_2 = 0 + 0.2(0.1) = 0.02$.
- **Step 3 (x = 0.2 to 0.3):** (0.2, 0.02). Slope $m_2 = 2(0.2) = 0.4$. $y_3 = 0.02 + 0.4(0.1) = 0.02 + 0.04 = 0.06$.
- **Conclusion:** The 3-step Euler approximation is 0.06 (note: the exact value is $x^2 = 0.3^2 = 0.09$).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** 0.09 is the exact solution $y(0.3) = (0.3)^2 = 0.09$, not the Euler approximation.
✗ **(C)** Stops after 2 steps and reports 0.04 or miscalculates 0.02 + 0.02.
✗ **(D)** Uses average of slopes instead of sequential Euler stepping.

Question 29 Analysis

★★★ MANDATORY

Correct Answer: (C) 4

STEP-BY-STEP SOLUTION

- **Step 1 (x = 0 to 1):** (0, 2). Slope $m_0 = 4 - 2 = 2$. $y_1 = 2 + 2(1) = 4$.
- **Step 2 (x = 1 to 2):** (1, 4). Slope $m_1 = 4 - 4 = 0$. $y_2 = 4 + 0(1) = 4$.
- **Conclusion:** The approximation for f(2) is 4.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes no net change from the initial value 2.
✗ **(B)** Averages 2 and 4 to get 3.
✗ **(D)** Adds the first slope twice: 2 + 2 + 2 = 6.

Question 30 Analysis

★★★ IMPORTANT

Correct Answer: (D) The approximation converges to the exact value of the particular solution

STEP-BY-STEP SOLUTION

- **Theoretical Basis of Euler's Method:** Euler's method is a first-order numerical integration method.
- **Limit Behavior:** As step size $\Delta x \to 0$, the sum of tangential line segments approaches the true integral of the slope field along the solution trajectory.
- **Conclusion:** Under standard continuity conditions, the sequence of Euler approximations converges to the exact analytical value of the solution.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Euler's method is numerically stable for well-behaved continuous differential equations, so it does not diverge.
✗ **(B)** Euler's approximation changes and improves with smaller step sizes.
✗ **(C)** Mathematically, the limit exists and converges.

Question 31 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $y = C e^{x^2}$

STEP-BY-STEP SOLUTION

- **Separate the Variables:** Divide both sides by y and multiply by dx: $(1/y)\,dy = 2x\,dx$.
- **Integrate Both Sides:** $\int (1/y)\,dy = \int 2x\,dx \Rightarrow \ln|y| = x^2 + C_1$.
- **Exponentiate Both Sides:** $|y| = e^{x^2 + C_1} = e^{C_1} \cdot e^{x^2}$.
- **Combine Constant:** Letting $C = \pm e^{C_1}$ (and including the equilibrium solution $y = 0$ when $C = 0$) yields $y = C e^{x^2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Incorrectly treats the constant of integration as additive after exponentiation: $e^{x^2} + C$ instead of $C e^{x^2}$.
✗ **(B)** Fails to integrate $(1/y)\,dy$ properly.
✗ **(D)** Integrates $2x$ as $2x$ instead of x^2.

Question 32 Analysis

★★★★ MANDATORY

Correct Answer: (A) $y = (x^2/4 + C)^2$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y^{(-1/2)}\,dy = x\,dx$.
- **Integrate Both Sides:** $\int y^{(-1/2)}\,dy = \int x\,dx \Rightarrow 2y^{(1/2)} = x^2/2 + C_1$.
- **Isolate √y:** $y^{(1/2)} = x^2/4 + C_1/2 = x^2/4 + C$.
- **Square Both Sides:** $y = (x^2/4 + C)^2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to divide by the coefficient 2 from the antiderivative $2\sqrt{y}$.
✗ **(C)** Squares each term individually instead of squaring the entire binomial sum $(x^2/4 + C)$.
✗ **(D)** Omits the constant C before squaring.

Question 33 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $y = 1 / (x^2 + C)$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $-y^{-2}\,dy = 2x\,dx$, or $(1/y^2)\,dy = -2x\,dx$.
- **Integrate Both Sides:** $\int y^{-2}\,dy = \int -2x\,dx \Rightarrow -y^{-1} = -x^2 - C \Rightarrow -1/y = -(x^2 + C)$.
- **Cancel Negative Signs:** $1/y = x^2 + C$.
- **Solve for y:** $y = 1 / (x^2 + C)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Fails to cancel the negative signs from both sides of the equation.
✗ **(B)** Mistakes a power of y for a logarithmic/exponential separation.
✗ **(C)** Integrates y^{-2} as a linear function.

Question 34 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $y = -\ln(C - e^x)$

STEP-BY-STEP SOLUTION

- **Exponent Laws:** Rewrite the right side as a product: $e^{x+y} = e^x \cdot e^y$.
- **Separate Variables:** Divide by e^y: $e^{-y}\,dy = e^x\,dx$.
- **Integrate Both Sides:** $\int e^{-y}\,dy = \int e^x\,dx \Rightarrow -e^{-y} = e^x + C_1$.
- **Isolate y:** $e^{-y} = -e^x - C_1 = C - e^x$ (where $C = -C_1$). Taking natural log: $-y = \ln(C - e^x) \Rightarrow y = -\ln(C - e^x)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the negative sign from the antiderivative of e^{-y}.
✗ **(C)** Attempts an invalid additive separation.
✗ **(D)** Sign error on the exponential term inside the logarithm.

Question 35 Analysis

★★★★ MANDATORY

Correct Answer: (A) $y^2 = \sin x + C$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $2y\,dy = \cos x\,dx$.
- **Integrate Both Sides:** $\int 2y\,dy = \int \cos x\,dx \Rightarrow y^2 = \sin x + C$.
- **Conclusion:** The general solution is $y^2 = \sin x + C$ (or $y = \sqrt{(\sin x + C)}$ for $y > 0$).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Integrates $\cos x$ to $-\sin x$ instead of $+\sin x$.
✗ **(C)** Forgets to integrate $2y$ to y^2.
✗ **(D)** Fails to simplify the antiderivative $\int 2y\,dy = y^2$.

Question 36 Analysis

★★★★ MANDATORY

Correct Answer: (C) $y^2 = x^2 + 2x + C$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y\,dy = (x + 1)\,dx$.
- **Integrate Both Sides:** $\int y\,dy = \int (x + 1)\,dx \Rightarrow y^2/2 = x^2/2 + x + C_1$.
- **Clear the Fraction:** Multiply the entire equation by 2: $y^2 = x^2 + 2x + 2C_1 = x^2 + 2x + C$.
- **Conclusion:** The general solution is $y^2 = x^2 + 2x + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Treats the derivative as an algebraic fraction and cross-multiplies incorrectly.
✗ **(B)** Integrates $(x + 1)$ to $(x + 1)^2$ without the factor 1/2.
✗ **(D)** Forgets to multiply the right side by 2 when clearing the denominator on $y^2/2$.

Question 37 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $y = C e^{x + x^3/3}$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $(1/y)\,dy = (1 + x^2)\,dx$.
- **Integrate Both Sides:** $\int (1/y)\,dy = \int (1 + x^2)\,dx \Rightarrow \ln(y) = x + x^3/3 + C_1$.
- **Exponentiate Both Sides:** $y = e^{x + x^3/3 + C_1} = e^{C_1} \cdot e^{x + x^3/3}$.
- **Absorb Constant:** Letting $C = e^{C_1}$ yields $y = C e^{x + x^3/3}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Treats the integration constant as additive outside the exponential function.
✗ **(B)** Omits the exponential step entirely.
✗ **(C)** Differentiates $1 + x^2$ to $2x$ instead of integrating.

Question 38 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $y = C e^{\tan x}$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $(1/y)\,dy = \sec^2 x\,dx$.
- **Integrate Both Sides:** $\int (1/y)\,dy = \int \sec^2 x\,dx \Rightarrow \ln(y) = \tan x + C_1$.
- **Exponentiate:** $y = e^{\tan x + C_1} = C e^{\tan x}$.
- **Conclusion:** $y = C e^{\tan x}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the logarithm and exponential transformation.
✗ **(C)** Mistakes $\sec x$ for the antiderivative of $\sec^2 x$.
✗ **(D)** Takes the logarithm of $\tan x$.

Question 39 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) Exponentiating both sides yields $e^{4x + C} = e^{4x} \cdot e^{C} = A e^{4x}$, which produces a multiplicative constant rather than an additive constant + C

STEP-BY-STEP SOLUTION

- **Rules of Exponents:** $e^{a+b} = e^a \cdot e^b$.
- **Application to $\ln|y| = 4x + C$:** Exponentiating both sides gives $|y| = e^{4x + C} = e^{4x} \cdot e^{C}$.
- **Multiplicative Constant:** e^C is a positive constant A (or arbitrary constant $C = \pm e^C$), making the correct form $y = C e^{4x}$.
- **Student's Error:** The student mistakenly treated the constant as additive (+ C) rather than multiplicative (· C).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** The derivative of $\ln|y|$ is indeed $1/y$.
✗ **(C)** C is an arbitrary real constant determined by initial conditions, not necessarily zero.
✗ **(D)** $dy/dx = 4y$ is separable and standard.

Question 40 Analysis

★★★★ MANDATORY

Correct Answer: (D) $y = 1 + C e^{3x}$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $(1 / (y - 1))\, dy = 3\, dx$.
- **Integrate Both Sides:** $\int (1 / (y - 1))\, dy = \int 3\, dx \Rightarrow \ln|y - 1| = 3x + C_1$.
- **Exponentiate:** $|y - 1| = e^{3x + C_1} = e^{C_1} \cdot e^{3x} \Rightarrow y - 1 = C e^{3x}$.
- **Solve for y:** $y = 1 + C e^{3x}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error when isolating y: subtracts 1 instead of adding 1.
✗ **(B)** Places the constant 1 in the exponent.
✗ **(C)** Integrates $1/(y - 1)$ as a linear term.

Question 41 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $y = 1 + 2 e^{x^2}$

STEP-BY-STEP SOLUTION

- **Separate the Variables:** Divide by $(y - 1)$ and multiply by dx: $(1 / (y - 1))\, dy = 2x\, dx$.
- **Integrate Both Sides:** $\int (1 / (y - 1))\, dy = \int 2x\, dx \Rightarrow \ln|y - 1| = x^2 + C$.
- **Apply Initial Condition y(0) = 3:** $\ln|3 - 1| = 0^2 + C \Rightarrow \ln(2) = C$.
- **Substitute C and Exponentiate:** $\ln|y - 1| = x^2 + \ln(2) \Rightarrow |y - 1| = e^{x^2 + \ln 2} = 2 e^{x^2}$.
- **Solve for y:** Since $y(0) = 3 > 1$, we drop absolute values: $y - 1 = 2 e^{x^2} \Rightarrow y = 1 + 2 e^{x^2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Incorrectly uses the initial y-value 3 as the multiplying constant instead of $(y(0) - 1) = 2$.
✗ **(C)** Ignores the vertical shift $(y - 1)$ and separates as $dy/y = 2x\, dx$.
✗ **(D)** Antidifferentiates $2x$ incorrectly as $2x$ instead of x^2.

Question 42 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $y = 2 / (2 - x^2)$

STEP-BY-STEP SOLUTION

- **Separate the Variables:** $y^{-2}\, dy = x\, dx$.
- **Integrate Both Sides:** $\int y^{-2}\, dy = \int x\, dx \Rightarrow -y^{-1} = x^2/2 + C$, so $-1/y = x^2/2 + C$.
- **Apply Initial Condition f(0) = 1:** $-1/1 = 0/2 + C \Rightarrow C = -1$.
- **Substitute C:** $-1/y = x^2/2 - 1 = (x^2 - 2) / 2$.
- **Solve for y:** Multiply by -1 and invert: $1/y = (2 - x^2) / 2 \Rightarrow y = 2 / (2 - x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the factor of 1/2 when integrating $x\, dx$ to $x^2/2$.
✗ **(B)** Fails to distribute the negative sign properly when simplifying $(x^2 - 2)/2$.
✗ **(C)** Sign error when moving the constant C across the equality.

Question 43 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $y = -\sqrt{(x^2 + 8)}$

STEP-BY-STEP SOLUTION

- **Separate Variables:** Multiply both sides by y dx: $y\,dy = x\,dx$.
- **Integrate Both Sides:** $\int y\,dy = \int x\,dx \Rightarrow y^2/2 = x^2/2 + C_1 \Rightarrow y^2 = x^2 + C$ (where $C = 2C_1$).
- **Apply Initial Condition y(1) = −3:** $(-3)^2 = 1^2 + C \Rightarrow 9 = 1 + C \Rightarrow C = 8$.
- **Substitute C:** $y^2 = x^2 + 8$.
- **Select the Correct Branch:** Taking square roots gives $y = \pm\sqrt{(x^2 + 8)}$. Since $y(1) = -3 < 0$, we must choose the negative square root branch: $y = -\sqrt{(x^2 + 8)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Classic AP trap: overlooks the negative sign required to satisfy the negative initial value $y(1) = -3$.

✗ **(B)** Arithmetic error in evaluating $(-3)^2 = 9$, using $9 + 1 = 10$ instead of $9 - 1 = 8$.

✗ **(D)** Incorrectly simplifies $\sqrt{(x^2 + 8)}$ as a linear term (x + constant).

Question 44 Analysis

★★★★ MANDATORY

Correct Answer: (A) $y = \sqrt{((2/3)x^3 + 2x + 4)}$

STEP-BY-STEP SOLUTION

- **Separate the Variables:** $y\,dy = (x^2 + 1)\,dx$.
- **Integrate Both Sides:** $\int y\,dy = \int (x^2 + 1)\,dx \Rightarrow y^2/2 = x^3/3 + x + C$.
- **Apply Initial Condition y(0) = 2:** $2^2/2 = 0 + 0 + C \Rightarrow 4/2 = 2 \Rightarrow C = 2$.
- **Isolate y^2:** $y^2/2 = x^3/3 + x + 2 \Rightarrow y^2 = 2(x^3/3 + x + 2) = (2/3)x^3 + 2x + 4$.
- **Solve for y:** Since $y(0) = 2 > 0$, choose the positive branch: $y = \sqrt{((2/3)x^3 + 2x + 4)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to multiply the entire right-hand side by 2 when clearing the denominator in $y^2/2$.

✗ **(C)** Multiplies only the polynomial terms by 2 while forgetting to double the constant C.

✗ **(D)** Forgets to take the square root of y^2.

Question 45 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $y = 5\,e^{\sin x}$

STEP-BY-STEP SOLUTION

- **Separate the Variables:** $(1/y)\,dy = \cos x\,dx$.
- **Integrate Both Sides:** $\int (1/y)\,dy = \int \cos x\,dx \Rightarrow \ln|y| = \sin x + C$.
- **Apply Initial Condition y(0) = 5:** $\ln(5) = \sin(0) + C = 0 + C \Rightarrow C = \ln 5$.
- **Exponentiate:** $\ln|y| = \sin x + \ln 5 \Rightarrow y = e^{\sin x + \ln 5} = e^{\ln 5} \cdot e^{\sin x} = 5\,e^{\sin x}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Incorrectly treats the constant of integration as additive after exponentiation: $e^{\sin x} + C$ with $C = 4$.

✗ **(C)** Signs error in antiderivative of cos x (antiderivative of cos x is positive sin x, not −sin x).

✗ **(D)** Confuses exponential separation with a direct integration of $dy/dx = \cos x$.

Question 46 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $y = (-1 + \sqrt{(4x^3 + 9)}) / 2$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $(2y + 1)\,dy = 3x^2\,dx$.
- **Integrate Both Sides:** $\int (2y + 1)\,dy = \int 3x^2\,dx \Rightarrow y^2 + y = x^3 + C$.
- **Apply Initial Condition y(0) = 1:** $1^2 + 1 = 0^3 + C \Rightarrow C = 2$.
- **Set up Quadratic in y:** $y^2 + y - (x^3 + 2) = 0$.
- **Apply Quadratic Formula:** $y = [-1 \pm \sqrt{(1^2 - 4(1)(-(x^3 + 2)))}] / 2 = [-1 \pm \sqrt{(1 + 4x^3 + 8)}] / 2 = [-1 \pm \sqrt{(4x^3 + 9)}] / 2$.
- **Select Branch for y(0) = 1:** For $x = 0$, $y = (-1 + \sqrt{9})/2 = (-1 + 3)/2 = 1$. Thus, $y = (-1 + \sqrt{(4x^3 + 9)}) / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Algebra error in calculating the discriminant: $1 + 4 = 5$ instead of $1 + 8 = 9$.
✗ **(B)** Sign error in the quadratic formula: uses $+1$ instead of $-b = -1$.
✗ **(D)** Incorrectly simplifies $(2y + 1)\,dy$.

Question 47 Analysis

★★★★ MANDATORY

Correct Answer: (B) y = (3e^x – 2)^(1/3)

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y^2\,dy = e^x\,dx$.
- **Integrate Both Sides:** $\int y^2\,dy = \int e^x\,dx \Rightarrow y^3/3 = e^x + C$.
- **Apply Initial Condition y(0) = 1:** $1^3/3 = e^0 + C \Rightarrow 1/3 = 1 + C \Rightarrow C = -2/3$.
- **Substitute C:** $y^3/3 = e^x - 2/3 \Rightarrow y^3 = 3(e^x - 2/3) = 3e^x - 2$.
- **Solve for y:** Take the cube root: $y = (3e^x - 2)^{(1/3)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Ignores the constant of integration completely.
✗ **(C)** Mistake in computing C: $1/3 - 1$ evaluated as $+1/3$ instead of $-2/3$.
✗ **(D)** Forgets to take the cube root of y^3 at the end.

Question 48 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $y = e^{x^2 + x - 1}$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $(1/y)\,dy = (2x + 1)\,dx$.
- **Integrate Both Sides:** $\int (1/y)\,dy = \int (2x + 1)\,dx \Rightarrow \ln|y| = x^2 + x + C$.
- **Apply Initial Condition y(1) = e:** $\ln(e) = 1^2 + 1 + C \Rightarrow 1 = 2 + C \Rightarrow C = -1$.
- **Exponentiate Both Sides:** $\ln y = x^2 + x - 1 \Rightarrow y = e^{x^2 + x - 1}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes $C = 0$ without evaluating the initial condition properly.
✗ **(B)** Arithmetic error in solving $1 = 2 + C$, getting $C = -2$.
✗ **(C)** Multiplies by the initial value e rather than solving for the true exponent constant C.

Question 49 Analysis

★★★★ MANDATORY

Correct Answer: (A) $y = 1 / (\ln(x + 1) + 1)$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $-y^{-2}\,dy = (1 / (x + 1))\,dx$, or $y^{-2}\,dy = -(1 / (x + 1))\,dx$.
- **Integrate Both Sides:** $\int -y^{-2}\,dy = \int (1 / (x + 1))\,dx \Rightarrow 1/y = \ln|x + 1| + C$.
- **Apply Initial Condition f(0) = 1:** $1/1 = \ln|0 + 1| + C \Rightarrow 1 = \ln(1) + C = 0 + C \Rightarrow C = 1$.
- **Substitute C:** $1/y = \ln(x + 1) + 1$ (for $x > -1$, $|x + 1| = x + 1$).
- **Solve for y:** Invert both sides: $y = 1 / (\ln(x + 1) + 1)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error with the antiderivative of $-y^{-2}$, introducing an erroneous negative sign into the denominator.

✗ **(C)** Carries an extra negative sign in the numerator.

✗ **(D)** Confuses polynomial/rational integration with exponential separation.

Question 50 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $y = 4 - 1 / (x + 1)$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $(y - 4)^{-2}\,dy = dx$.
- **Integrate Both Sides:** $\int (y - 4)^{-2}\,dy = \int dx \Rightarrow -(y - 4)^{-1} = x + C \Rightarrow -1 / (y - 4) = x + C$.
- **Apply Initial Condition y(0) = 3:** $-1 / (3 - 4) = 0 + C \Rightarrow -1 / (-1) = 1 = C$.
- **Substitute C:** $-1 / (y - 4) = x + 1$.
- **Solve for y:** Multiply by -1: $1 / (y - 4) = -(x + 1) \Rightarrow y - 4 = -1 / (x + 1) \Rightarrow y = 4 - 1 / (x + 1)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error when reciprocating: drops the negative sign from $-1 / (x + 1)$.

✗ **(B)** Sign error when solving for C, using $C = -1$ instead of $+1$.

✗ **(D)** Mistakes $(y - 4)^2$ for y^2 and ignores the vertical shift.

Question 51 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $y = 1 / (1 - x^2)$ on $(-1, 1)$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y^{-2}\,dy = 2x\,dx$.
- **Integrate Both Sides:** $\int y^{-2}\,dy = \int 2x\,dx \Rightarrow -1/y = x^2 + C$.
- **Apply Initial Condition y(0) = 1:** $-1/1 = 0^2 + C \Rightarrow C = -1$.
- **Solve for y:** $-1/y = x^2 - 1 \Rightarrow 1/y = 1 - x^2 \Rightarrow y = 1 / (1 - x^2)$.
- **Determine Domain:** The function y is undefined when $1 - x^2 = 0 \Rightarrow x = \pm 1$. The solution to a differential equation must be continuous and differentiable on an open interval containing the initial point $x = 0$. Therefore, the interval is $(-1, 1)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error in integrating or solving for C, arriving at $1 / (1 + x^2)$.

✗ **(C)** Selects an interval that does not contain the initial point $x = 0$.

✗ **(D)** Fails to multiply both sides by -1 when solving for y.

Question 52 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $y = -\sqrt{(x+4)}$ on $(-4, \infty)$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $2y\,dy = dx$.
- **Integrate Both Sides:** $\int 2y\,dy = \int dx \Rightarrow y^2 = x + C$.
- **Apply Initial Condition y(0) = −2:** $(-2)^2 = 0 + C \Rightarrow C = 4$.
- **Select Branch for y:** $y^2 = x + 4 \Rightarrow y = \pm\sqrt{(x+4)}$. Since $y(0) = -2 < 0$, $y = -\sqrt{(x+4)}$.
- **Determine Open Domain:** For $dy/dx = 1/(2y)$ to be defined, y cannot equal 0. Thus $x + 4 > 0 \Rightarrow x > -4$. The domain must be the open interval $(-4, \infty)$ because the derivative does not exist at $x = -4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Chooses the positive square root branch, violating the initial condition $y(0) = -2$.

✗ **(C)** Includes the boundary $x = -4$; at $x = -4$, $y = 0$, where dy/dx is undefined.

✗ **(D)** Arithmetic error in squaring −2, setting $(-2)^2 = 2$.

Question 53 Analysis

★★★★ MANDATORY

Correct Answer: (C) $y = 1 / (1 - \sin x)$ on $(-3\pi/2, \pi/2)$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y^{-2}\,dy = \cos x\,dx$.
- **Integrate Both Sides:** $\int y^{-2}\,dy = \int \cos x\,dx \Rightarrow -1/y = \sin x + C$.
- **Apply Initial Condition y(0) = 1:** $-1/1 = \sin(0) + C = 0 + C \Rightarrow C = -1$.
- **Solve for y:** $-1/y = \sin x - 1 \Rightarrow y = 1/(1 - \sin x)$.
- **Find Domain:** y is undefined when $1 - \sin x = 0 \Rightarrow \sin x = 1$. The closest values of x where $\sin x = 1$ on either side of 0 are $x = \pi/2$ and $x = -3\pi/2$. Thus the maximal open interval containing 0 is $(-3\pi/2, \pi/2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error when integrating or solving for C.

✗ **(B)** Confuses the antiderivative of $\cos x$ with $\sin x$.

✗ **(D)** Incorrectly assumes the left boundary where $\sin x = 1$ is $-\pi/2$ (at $-\pi/2$, $\sin(-\pi/2) = -1 \neq 1$).

Question 54 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $y = -\sqrt{((x-2)^2 + 1)}$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y\,dy = (x - 2)\,dx$.
- **Integrate Both Sides:** $\int y\,dy = \int (x-2)\,dx \Rightarrow y^2/2 = (x-2)^2/2 + C_1 \Rightarrow y^2 = (x-2)^2 + C$.
- **Apply Initial Condition y(2) = −1:** $(-1)^2 = (2-2)^2 + C \Rightarrow 1 = 0 + C \Rightarrow C = 1$.
- **Substitute C:** $y^2 = (x-2)^2 + 1$.
- **Choose Negative Branch:** Since $y(2) = -1 < 0$, $y = -\sqrt{((x-2)^2 + 1)}$. (Note: Defined for all $x \in \mathbb{R}$ since $(x-2)^2 + 1 \geq 1 > 0$).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Overlooks the negative initial condition value $y(2) = -1$.

✗ **(B)** Arithmetic error in expanding or computing the constant C.

✗ **(C)** Attempts linear approximation instead of finding the exact particular solution.

Question 55 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $x \in (0, \sqrt{e})$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y^{-2}\,dy = (1/x)\,dx$.
- **Integrate Both Sides:** $\int y^{-2}\,dy = \int (1/x)\,dx \Rightarrow -1/y = \ln|x| + C = \ln x + C$ (since $x > 0$).
- **Apply Initial Condition y(1) = 2:** $-1/2 = \ln(1) + C = 0 + C \Rightarrow C = -1/2$.
- **Substitute and Solve for y:** $-1/y = \ln x - 1/2 = (2\ln x - 1)/2 \Rightarrow y = 2/(1 - 2\ln x)$.
- **Find Domain:** The differential equation requires $x > 0$. The denominator $1 - 2\ln x = 0$ when $2\ln x = 1 \Rightarrow \ln x = 1/2 \Rightarrow x = e^{1/2} = \sqrt{e} \approx 1.649$. The interval must contain the initial point $x = 1$. Since $0 < 1 < \sqrt{e}$, the domain of validity is $(0, \sqrt{e})$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Overlooks the vertical asymptote where the denominator becomes zero.

✗ **(B)** Solves $1 - \ln x = 0$ instead of $1 - 2\ln x = 0$.

✗ **(D)** Selects the open interval to the right of the vertical asymptote, which does not contain the initial point $x = 1$.

Question 56 Analysis

★★★★ MANDATORY

Correct Answer: (D) $y = 1/(4 - x^4)$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y^{-2}\,dy = 4x^3\,dx$.
- **Integrate Both Sides:** $\int y^{-2}\,dy = \int 4x^3\,dx \Rightarrow -1/y = x^4 + C$.
- **Apply Initial Condition y(0) = 1/4:** $-1/(1/4) = 0^4 + C \Rightarrow -4 = C$.
- **Substitute C:** $-1/y = x^4 - 4$.
- **Solve for y:** Multiply by -1 and invert: $1/y = 4 - x^4 \Rightarrow y = 1/(4 - x^4)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error when moving the constant C across the equation.

✗ **(B)** Inverts 1/4 incorrectly or introduces extraneous coefficients.

✗ **(C)** Confuses reciprocal power with direct integration.

Question 57 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $y = \sqrt{(2x + 7)}$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y\,dy = dx$.
- **Integrate Both Sides:** $\int y\,dy = \int dx \Rightarrow y^2/2 = x + C$.
- **Apply Initial Condition y(1) = 3:** $3^2/2 = 1 + C \Rightarrow 9/2 = 1 + C \Rightarrow C = 7/2$.
- **Solve for y^2:** $y^2/2 = x + 7/2 \Rightarrow y^2 = 2x + 7$.
- **Choose Branch:** Since $y(1) = 3 > 0$, $y = \sqrt{(2x + 7)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error in calculating $C = 9/2 - 1 = 7/2$, mistakenly using 8/2.

✗ **(B)** Forgets to multiply x and C by 2 when isolating y^2.

✗ **(D)** Linearizes the differential equation erroneously.

Question 58 Analysis

★★★★ MANDATORY

Correct Answer: (B) $y = -\sqrt{(2e^x + 2)}$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $y\,dy = e^x\,dx$.
- **Integrate Both Sides:** $\int y\,dy = \int e^x\,dx \Rightarrow y^2/2 = e^x + C$.
- **Apply Initial Condition y(0) = −2:** $(-2)^2/2 = e^0 + C \Rightarrow 4/2 = 1 + C \Rightarrow 2 = 1 + C \Rightarrow C = 1$.
- **Substitute C:** $y^2/2 = e^x + 1 \Rightarrow y^2 = 2e^x + 2$.
- **Choose Branch:** Because $y(0) = -2 < 0$, we must take the negative root: $y = -\sqrt{(2e^x + 2)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Fails to take the negative square root branch needed for $y(0) = -2$.
✗ **(C)** Does not divide $(-2)^2$ by 2 on the left-hand side, obtaining $C = 3$ and $2C = 6$.
✗ **(D)** Forgets to multiply e^x by 2 when isolating y^2.

Question 59 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $y = \tan(x^2/2)$ on $(-\sqrt{\pi}, \sqrt{\pi})$

STEP-BY-STEP SOLUTION

- **Separate Variables:** $(1 / (1 + y^2))\,dy = x\,dx$.
- **Integrate Both Sides:** $\int (1 / (1 + y^2))\,dy = \int x\,dx \Rightarrow \arctan(y) = x^2/2 + C$.
- **Apply Initial Condition y(0) = 0:** $\arctan(0) = 0^2/2 + C \Rightarrow 0 = C$.
- **Solve for y:** $\arctan(y) = x^2/2 \Rightarrow y = \tan(x^2/2)$.
- **Find Domain of Definition:** The tangent function $\tan(u)$ is defined on $(-\pi/2, \pi/2)$. Thus we require $-\pi/2 < x^2/2 < \pi/2 \Rightarrow -\pi < x^2 < \pi$. Since $x^2 \geq 0$, this simplifies to $x^2 < \pi \Rightarrow -\sqrt{\pi} < x < \sqrt{\pi}$. This open interval contains the initial point $x = 0$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrates $x\,dx$ as x^2 instead of $x^2/2$.
✗ **(C)** Overlooks the vertical asymptotes of the tangent function.
✗ **(D)** Inverts arctan incorrectly as arctan rather than tan.

Question 60 Analysis

★★★★ MANDATORY

Correct Answer: (D) y = (x²+ 7)^(1/3)

STEP-BY-STEP SOLUTION

- **Separate Variables:** $3y^2\,dy = 2x\,dx$.
- **Integrate Both Sides:** $\int 3y^2\,dy = \int 2x\,dx \Rightarrow y^3 = x^2 + C$.
- **Apply Initial Condition y(1) = 2:** $2^3 = 1^2 + C \Rightarrow 8 = 1 + C \Rightarrow C = 7$.
- **Substitute C and Solve:** $y^3 = x^2 + 7 \Rightarrow y = (x^2 + 7)^{(1/3)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error in finding C: 8 – 1 miscalculated as 8.
✗ **(B)** Introduces an erroneous factor of 2 during integration.
✗ **(C)** Cubes both sides instead of taking the cube root.

Question 61 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 13,500

STEP-BY-STEP SOLUTION

- **Identify the Model:** The differential equation $dy/dt = ky$ has general solution $y(t) = y(0)\,e^{kt}$.
- **Determine Growth Multiplier:** At $t = 2$, $y(2) = 500\,e^{2k} = 1{,}500 \Rightarrow e^{2k} = 1{,}500 / 500 = 3$. The population triples every 2 time units.
- **Calculate y(6):** Time $t = 6$ corresponds to $6/2 = 3$ tripling periods.
- **Compute Final Value:** $y(6) = y(0) \cdot (e^{2k})^3 = 500 \cdot (3)^3 = 500 \cdot 27 = 13{,}500$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Treats growth as linear: adding 1,000 every 2 units gives 500 + 3,000 = 3,500 or doubles it to 4,500.
✗ **(B)** Calculates 2 tripling periods instead of 3: $500 \cdot 9 = 4{,}500$ or multiplies 1,500 by 6.
✗ **(D)** Computes 4 tripling periods: $500 \cdot 81 = 40{,}500$.

Question 62 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 10 grams

STEP-BY-STEP SOLUTION

- **Understand Exponential Decay:** For $dM/dt = -k\,M$, the mass remaining after t days is $M(t) = M_0\,(1/2)^{(t / t_half)}$.
- **Identify Parameters:** Initial mass $M_0 = 80$ g, half-life $t_half = 10$ days, elapsed time $t = 30$ days.
- **Find Number of Half-Lives:** Number of half-lives $= 30 / 10 = 3$.
- **Compute Remaining Mass:** $M(30) = 80 \cdot (1/2)^3 = 80 \cdot (1/8) = 10$ grams.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates for 2 half-lives: $80 / 4 = 20$ grams.
✗ **(C)** Calculates for 4 half-lives: $80 / 16 = 5$ grams.
✗ **(D)** Calculates for 5 half-lives: $80 / 32 = 2.5$ grams.

Question 63 Analysis

★★★★ MANDATORY

Correct Answer: (D) k = (ln 2) / 4

STEP-BY-STEP SOLUTION

- **Model Equation:** $P(t) = P_0\,e^{kt}$.
- **Apply Doubling Condition:** At $t = 4$, $P(4) = 2\,P_0$.
- **Set up Exponential Equation:** $2\,P_0 = P_0\,e^{4k} \Rightarrow e^{4k} = 2$.
- **Solve for k:** Take natural logarithm of both sides: $4k = \ln 2 \Rightarrow k = (\ln 2) / 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplies by 4 instead of dividing: $k = 4 \ln 2$.
✗ **(B)** Inverts the relationship between base and time.
✗ **(C)** Mistakes exponential rate for a simple linear fraction 1/4.

Question 64 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $T(t) = 70 + 120\,(2/3)^{(t/10)}$

STEP-BY-STEP SOLUTION

- **Set up Differential Equation:** $dT/dt = -k(T - 70)$. Let $u = T - 70$, then $du/dt = -ku \Rightarrow u(t) = u_0\, e^{-kt}$.
- **Find Initial Difference u_0:** At $t = 0$, $u(0) = T(0) - 70 = 190 - 70 = 120°F$.
- **Apply t = 10 Condition:** At $t = 10$, $T(10) = 150 \Rightarrow u(10) = 150 - 70 = 80°F$.
- **Find Exponential Decay Base:** $120\, e^{-10k} = 80 \Rightarrow e^{-10k} = 80/120 = 2/3$.
- **Write Function T(t):** $u(t) = 120 \cdot (e^{-10k})^{(t/10)} = 120\,(2/3)^{(t/10)}$.
- **Reconstruct T(t):** $T(t) = 70 + u(t) = 70 + 120\,(2/3)^{(t/10)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses the initial temperature 190 as the coefficient of the decaying term rather than the temperature difference $190 - 70 = 120$.
✗ **(C)** Incorrectly subtracts the decay term from 190.
✗ **(D)** Miscalculates the ratio 80/120 as 3/4 instead of 2/3.

Question 65 Analysis

★★★★ MANDATORY

Correct Answer: (B) $t = 40 \ln 2$

STEP-BY-STEP SOLUTION

- **Write Solution Formula:** For $dy/dt = -0.05\,y$, $y(t) = y(0)\, e^{-0.05\,t} = 200\, e^{-0.05\,t}$.
- **Set Target Value:** $50 = 200\, e^{-0.05\,t}$.
- **Isolate Exponential:** $e^{-0.05\,t} = 50/200 = 1/4$.
- **Take Natural Logarithm:** $-0.05\,t = \ln(1/4) = -\ln(4) = -2\ln(2)$.
- **Solve for t:** $t = (-2\ln 2)/(-0.05) = (2\ln 2)/(1/20) = 40 \ln 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the power 2 from $\ln(4) = 2\ln(2)$: computes $(\ln 2)/0.05 = 20 \ln 2$.
✗ **(C)** Miscalculates $1/0.05$ as 10 rather than 20.
✗ **(D)** Arithmetic error in dividing 2 by 0.05.

Question 66 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $r = (\ln 3)/20$

STEP-BY-STEP SOLUTION

- **General Solution:** $A(t) = A_0\, e^{rt}$.
- **Apply Tripling Condition:** At $t = 20$, $A(20) = 3\,A_0$.
- **Equate and Cancel A_0:** $3\,A_0 = A_0\, e^{20r} \Rightarrow e^{20r} = 3$.
- **Solve for r:** Take natural logarithms: $20r = \ln 3 \Rightarrow r = (\ln 3)/20$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Swaps the arguments of log and the denominator: $(\ln 20)/3$.
✗ **(C)** Multiplies by 20 instead of dividing.
✗ **(D)** Assumes simple interest without compounding: $3/20$.

Question 67 Analysis

★★★★ MANDATORY

Correct Answer: (C) 64%

STEP-BY-STEP SOLUTION

- **Understand Remaining Fraction:** If 20% decomposes in 5 hours, then 80% (or a fraction of 0.80) remains after 5 hours.
- **Model Representation:** $M(5) / M_0 = e^{-5k} = 0.80$.
- **Evaluate at t = 10 hours:** 10 hours consists of two 5–hour intervals.
- **Calculate Fraction Remaining:** $M(10) / M_0 = (e^{-5k})^2 = (0.80)^2 = 0.64 = 64\%$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Classic linear misconception: assumes another 20% decomposes linearly, yielding 100% – 40% = 60%.
✗ **(B)** Doubles the decomposed percentage to 40% remaining.
✗ **(D)** Miscalculates $(0.80)^2$.

Question 68 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 20 + 80 / e

STEP-BY-STEP SOLUTION

- **Separate and Integrate:** $dT / (T - 20) = -0.1\,dt \Rightarrow \ln|T - 20| = -0.1\,t + C$.
- **Apply Initial Condition T(0) = 100:** $\ln|100 - 20| = \ln(80) = C$.
- **Express Temperature Difference:** $T(t) - 20 = 80\,e^{-0.1t} \Rightarrow T(t) = 20 + 80\,e^{-0.1t}$.
- **Evaluate at t = 10:** $T(10) = 20 + 80\,e^{-0.1 \cdot 10} = 20 + 80\,e^{-1} = 20 + 80 / e$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses 100 instead of the temperature differential 100 – 20 = 80.
✗ **(C)** Forgets to add the ambient temperature 20°C back at the end.
✗ **(D)** Signs error in exponent: uses e^{+1} instead of e^{-1}.

Question 69 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 3

STEP-BY-STEP SOLUTION

- **Model Equations:** $y(1) = y(0)\,e^k = 6$ and $y(3) = y(0)\,e^{3k} = 24$.
- **Divide Equations:** $y(3) / y(1) = [y(0)\,e^{3k}] / [y(0)\,e^k] = e^{2k} = 24 / 6 = 4$.
- **Find Factor e^k:** Since $e^{2k} = 4$ and $e^k > 0$, we have $e^k = \sqrt{4} = 2$.
- **Solve for y(0):** From $y(1) = y(0)\,e^k = 6 \Rightarrow y(0) \cdot 2 = 6 \Rightarrow y(0) = 6 / 2 = 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides 3 by 2 instead of 6 by 2.
✗ **(C)** Mistakes the ratio 24/6 = 4 for the initial value.
✗ **(D)** Mistakes the multiplier $e^k = 2$ for the initial value.

Question 70 Analysis

★★★★ MANDATORY

Correct Answer: (D) 4 years

STEP-BY-STEP SOLUTION

- **Relate Quadrupling and Doubling:** Quadrupling means multiplying by 4. Since $4 = 2^2$, quadrupling represents exactly 2 doublings.
- **Compute Doubling Time:** If 2 doubling periods take 8 years, each doubling period takes $8 / 2 = 4$ years.
- **Algebraic Verification:** $e^{8k} = 4 = 2^2 \Rightarrow (e^{4k})^2 = 2^2 \Rightarrow e^{4k} = 2$. Thus the population doubles every 4 years.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides 8 by 4 instead of by 2.
✗ **(B)** Multiplies 8 by 2 to get 16 years.
✗ **(C)** Arithmetic error or guesswork.

Question 71 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) Carrying capacity = 800; growing most rapidly when P = 400

STEP-BY-STEP SOLUTION

- **Identify the Standard Logistic Form:** The equation $dP/dt = k P (1 - P / M)$ has carrying capacity M.
- **Determine Carrying Capacity M:** Matching with $dP/dt = 0.04 P (1 - P / 800)$, we see $M = 800$.
- **Find Population for Maximum Growth Rate:** For any logistic model, the growth rate dP/dt is a downward-opening parabola with respect to P, reaching its absolute maximum at the vertex: $P = M / 2$.
- **Compute Optimal Size:** $P = 800 / 2 = 400$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Mistakes the carrying capacity for the point of maximum growth; at $P = 800$, the growth rate is 0.
✗ **(B)** Confuses M/2 with carrying capacity.
✗ **(D)** Doubles the carrying capacity erroneously.

Question 72 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1,500

STEP-BY-STEP SOLUTION

- **Factor into Standard Form:** Factor out 3P from both terms: $dP/dt = 3P (1 - P / (3 \cdot 500)) = 3P (1 - P / 1{,}500)$.
- **Equate Carrying Capacity:** In the form $dP/dt = kP(1 - P/M)$, the denominator inside the parenthesis is the carrying capacity M.
- **Alternative Equilibrium Method:** Set $dP/dt = 0$: $3P - P^2 / 500 = 0 \Rightarrow P(3 - P/500) = 0$.
- **Find Non-zero Steady State:** $3 - P / 500 = 0 \Rightarrow P = 3 \cdot 500 = 1{,}500$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Overlooks the coefficient 3 and assumes the denominator 500 is directly the carrying capacity.
✗ **(B)** Divides 1,500 by 2, mistaking the point of maximum growth for carrying capacity.
✗ **(D)** Doubles the correct value.

Question 73 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 180

STEP-BY-STEP SOLUTION

- **Identify Carrying Capacity:** The equilibrium populations are $P = 0$ and $P = 600$, so $M = 600$.
- **Find Point of Maximum Growth Rate:** Maximum growth rate occurs at half the carrying capacity: $P = 600 / 2 = 300$.
- **Evaluate Maximum Rate:** Substitute $P = 300$ into the differential equation:
- $(dP/dt)_{max} = 0.002 \cdot (300) \cdot (600 - 300)$
- $(dP/dt)_{max} = 0.002 \cdot 300 \cdot 300 = 0.002 \cdot 90{,}000 = 180$ individuals/year.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Identifies the population $P = 300$ where the maximum rate occurs rather than computing the rate itself.

✗ **(C)** Calculates $0.002 \cdot 600 \cdot 300 = 360$, forgetting that $(600 - P) = 300$.

✗ **(D)** Divides the rate by 2 erroneously.

Question 74 Analysis

★★★★ MANDATORY

Correct Answer: (C) The graph is concave up initially, has a point of inflection when $P = 100$, and is concave down for $P > 100$.

STEP-BY-STEP SOLUTION

- **Identify Critical Values:** Carrying capacity $M = 200$, inflection point occurs at $P = M / 2 = 100$.
- **Analyze Initial Condition:** $P(0) = 50$, which is strictly less than 100.
- **Determine Concavity for $P < 100$:** For $0 < P < 100$, the rate of change dP/dt is increasing, which means $d^2P/dt^2 > 0$ (concave up).
- **Determine Concavity for $P > 100$:** For $100 < P < 200$, the rate of change dP/dt is decreasing toward 0, which means $d^2P/dt^2 < 0$ (concave down).
- **Conclusion:** The curve starts concave up, undergoes an inflection point at $P = 100$, and stays concave down as it approaches 200.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes logistic curves are always concave down (only true if $P(0) \geq M/2$).

✗ **(B)** Assumes population curves never change concavity.

✗ **(D)** Reverses the concavity before and after the inflection point.

Question 75 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\lim_{t \to \infty} P(t) = 1{,}200$; concave up

STEP-BY-STEP SOLUTION

- **Evaluate Limit as $t \to \infty$:** For any initial condition $P(0) > 0$, the logistic solution approaches the stable carrying capacity $M = 1{,}200$.
- **Determine First Derivative Sign:** Since $P(0) = 1{,}500 > 1{,}200$, $(1 - P / 1{,}200) < 0$, so $dP/dt < 0$ (population is decreasing).
- **Differentiate to Find d^2P/dt^2:**
- $dP/dt = 0.05\,P - (0.05 / 1{,}200)\,P^2$.
- $d^2P/dt^2 = (0.05 - (0.10 / 1{,}200)\,P) \cdot (dP/dt) = 0.05\,(1 - 2P / 1{,}200) \cdot (dP/dt)$.
- **Analyze Sign of Second Derivative:**
- Since $P > 1{,}200$, $2P / 1{,}200 > 2 > 1$, so $(1 - 2P / 1{,}200) < 0$.
- Since $dP/dt < 0$, we have $d^2P/dt^2 = (\text{negative}) \cdot (\text{negative}) > 0$.
- **Conclusion:** Because $d^2P/dt^2 > 0$, the curve is decreasing and concave up toward the horizontal asymptote $P = 1{,}200$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Intuitive guess that decreasing towards an asymptote must be concave down; forgets that a horizontal asymptote approached from above is concave up.

✗ **(C)** Believes the population stays at its initial value.

✗ **(D)** Mistakes carrying capacity for extinction.

Question 76 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) P = 5,000

STEP-BY-STEP SOLUTION

- **Identify Carrying Capacity:** From $dP/dt = kP(10{,}000 - P)$, the positive equilibrium carrying capacity is $M = 10{,}000$.
- **Recall Logistic Theorem:** The maximum rate of change of a logistic population always occurs at exactly half of the carrying capacity: $P = M/2$.
- **Compute Population:** $P = 10{,}000/2 = 5{,}000$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Selects the initial population value $P(0) = 1{,}000$.
✗ **(C)** Selects the carrying capacity $M = 10{,}000$, where growth rate is actually zero.
✗ **(D)** Divides carrying capacity by 4 instead of 2.

Question 77 Analysis

★★★★ MANDATORY

Correct Answer: (A) $4y(1 - y/25)(1 - y/50)$

STEP-BY-STEP SOLUTION

- **Expand First Derivative:** $dy/dt = 2y - 2y^2/50 = 2y - y^2/25$.
- **Differentiate with Respect to t using Chain Rule:**
- • $d^2y/dt^2 = d/dt[2y - y^2/25] = (2 - 2y/25) \cdot (dy/dt)$.
- **Factor Derivative Coefficient:** $2 - 2y/25 = 2(1 - y/25)$.
- **Substitute dy/dt:**
- • $d^2y/dt^2 = 2(1 - y/25) \cdot [2y(1 - y/50)] = 4y(1 - y/25)(1 - y/50)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Missing the factor of 2y from substituting dy/dt.
✗ **(C)** Omits the $(1 - y/25)$ factor resulting from differentiating with respect to y.
✗ **(D)** Forgets to multiply by dy/dt via the chain rule.

Question 78 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 0.20

STEP-BY-STEP SOLUTION

- **Recall Formula for Maximum Rate:** Maximum growth occurs at $P = M/2 = 1{,}200$.
- **Substitute P = M/2 into Logistic Formula:**
- • $(dP/dt)_{max} = k \cdot (M/2) \cdot (1 - (M/2)/M) = k \cdot (M/2) \cdot (1/2) = k \cdot M/4$.
- **Set Equal to Given Maximum Rate:**
- • $k \cdot (2{,}400)/4 = 120 \Rightarrow 600k = 120$.
- **Solve for k:** $k = 120/600 = 1/5 = 0.20$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates $120/2{,}400 = 0.05$, forgetting to account for the factor of 4 from $(M/2) \cdot (1/2)$.
✗ **(B)** Calculates $120/1{,}200 = 0.10$, forgetting the second factor of 1/2.
✗ **(D)** Multiplies by 2 instead of dividing.

Question 79 Analysis

★★★★ MANDATORY

Correct Answer: (A) $P(t) = 500 / (1 + 4e^{-0.1t})$

STEP-BY-STEP SOLUTION

- **General Solution to Logistic Differential Equation:** $P(t) = M / (1 + Ae^{-kt})$, where $M = 500$ and $k = 0.1$.
- **Apply Initial Condition P(0) = 100:**
- $100 = 500 / (1 + Ae^{0}) = 500 / (1 + A)$.
- **Solve for A:** $1 + A = 500 / 100 = 5 \Rightarrow A = 4$.
- **Construct Final Formula:** $P(t) = 500 / (1 + 4e^{-0.1t})$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sets $A = 5$ instead of $A = 5 - 1 = 4$.

✗ **(C)** Uses the initial population 100 in the numerator instead of the carrying capacity 500.

✗ **(D)** Sign error in the exponent: uses $e^{+0.1t}$, which would cause $P(t) \to 0$ as $t \to \infty$.

Question 80 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) The population grows fastest when P = 250.

STEP-BY-STEP SOLUTION

- **Analyze Statement A:** Carrying capacity $M = 250$, so inflection point is at $P = 125$. If $P(0) = 200 > 125$, the population is in the upper half of the curve, where $d^2P/dt^2 < 0$ for all $t \geq 0$. (TRUE).
- **Analyze Statement B:** If $P(0) = 50 < 125$, $P(t)$ increases through $P = 125$, where $d^2P/dt^2 = 0$, representing a point of inflection. (TRUE).
- **Analyze Statement C:** Carrying capacity $M = 250$ is the stable equilibrium for all $P(0) > 0$. (TRUE).
- **Analyze Statement D:** When $P = 250$, $dP/dt = 0.08(250)(1 - 250/250) = 0$ (growth rate is zero, not fastest). The population grows fastest when $P = M / 2 = 125$. (FALSE, hence D is the correct option).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Statement A is mathematically correct.

✗ **(B)** Statement B is mathematically correct.

✗ **(C)** Statement C is mathematically correct.

Question 81 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 5

STEP-BY-STEP SOLUTION

- **Find the First Derivative at (1, 2):**
- $dy/dx|_{(1,2)} = (1)(2) + 1 = 3$.
- **Differentiate with Respect to x using Product and Chain Rules:**
- $d^2y/dx^2 = d/dx[xy + 1] = (1)\cdot y + x\cdot(dy/dx) + 0 = y + x(dy/dx)$.
- **Substitute x = 1, y = 2, and dy/dx = 3:**
- $d^2y/dx^2|_{(1,2)} = 2 + (1)\cdot(3) = 2 + 3 = 5$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates only the first derivative $dy/dx = 3$, mistaking it for the second derivative.

✗ **(C)** Multiplies (1)(2)(3) or adds an extraneous term.

✗ **(D)** Omits the $x(dy/dx)$ term from the product rule.

Question 82 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2.8; underestimate because f''(1) > 0

STEP-BY-STEP SOLUTION

- **Find Slope at (1, 3):** $dy/dx|_{(1,3)} = 2(1) - 3 = -1$.
- **Write Tangent Line Equation:** $y - 3 = -1(x - 1) \Rightarrow y = 3 - (x - 1) = 4 - x$.
- **Approximate f(1.2):** $f(1.2) \approx 4 - 1.2 = 2.8$.
- **Calculate Second Derivative:**
- • $d^2y/dx^2 = d/dx\,[2x - y] = 2 - dy/dx$.
- • At (1, 3): $d^2y/dx^2 = 2 - (-1) = 3 > 0$.
- **Determine Over/Underestimate:** Since $d^2y/dx^2 > 0$ at and near $x = 1$, the curve is concave up, which means the tangent line lies entirely below the curve. Therefore, 2.8 is an underestimate.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Correct approximation value (2.8) but incorrectly claims an overestimate with $f''(1) < 0$.

✗ **(B)** Calculates the tangent slope as +1 instead of –1, yielding 3.2.

✗ **(D)** Incorrect approximation and incorrect concavity.

Question 83 Analysis

★★★★ MANDATORY

Correct Answer: (A) dS/dt = 2 – 0.04 S

STEP-BY-STEP SOLUTION

- **General Mixing Formula:** $dS/dt = \text{Rate_in} - \text{Rate_out}$.
- **Compute Rate_in:** $(0.5\ \text{kg/L}) \cdot (4\ \text{L/min}) = 2\ \text{kg/min}$.
- **Compute Rate_out:** $\text{Concentration_out} \cdot \text{Flow_rate_out} = (S(t)/100\ \text{kg/L}) \cdot (4\ \text{L/min}) = (4/100)\,S = 0.04\,S\ \text{kg/min}$.
- **Assemble Differential Equation:** $dS/dt = 2 - 0.04\,S$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses concentration (0.5) directly as rate of salt in, forgetting to multiply by the volumetric flow rate 4 L/min.

✗ **(C)** Uses the volumetric outflow rate 4 rather than the concentration fraction 4/100.

✗ **(D)** Uses 2/100 = 0.02 instead of 4/100 = 0.04.

Question 84 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) dS/dt = 10 – 3S / (500 + 2t)

STEP-BY-STEP SOLUTION

- **Find Volume Function V(t):** Inflow rate is 5 gal/min and outflow rate is 3 gal/min, so volume increases by 5 – 3 = 2 gal/min: $V(t) = 500 + 2t$ gallons.
- **Compute Salt Inflow Rate:** $\text{Rate_in} = (2\ \text{lbs/gal}) \cdot (5\ \text{gal/min}) = 10\ \text{lbs/min}$.
- **Compute Salt Outflow Rate:** Concentration at time t is $S(t)/V(t) = S(t)/(500 + 2t)$.
- • $\text{Rate_out} = [S(t)/(500 + 2t)] \cdot (3\ \text{gal/min}) = 3S/(500 + 2t)$.
- **Combine Terms:** $dS/dt = \text{Rate_in} - \text{Rate_out} = 10 - 3S/(500 + 2t)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes constant volume 500 gallons, ignoring that inflow exceeds outflow.

✗ **(C)** Uses the inflow rate 5 in the numerator of the outflow term instead of 3.

✗ **(D)** Fails to multiply concentration of inflow (2) by inflow rate (5).

Question 85 Analysis

★★★★ MANDATORY

Correct Answer: (A) $d^2y/dx^2 = (2y^3 - y^2) / x^2$

STEP-BY-STEP SOLUTION

- **Rewrite dy/dx for Differentiation:** $dy/dx = y^2 \cdot x^{-1}$.
- **Differentiate with Respect to x using Product Rule and Chain Rule:**
- • $d^2y/dx^2 = d/dx [y^2 \cdot x^{-1}] = [2y \cdot (dy/dx)] \cdot x^{-1} + y^2 \cdot (-x^{-2})$.
- **Substitute $dy/dx = y^2 / x$:**
- • $d^2y/dx^2 = 2y \cdot (y^2 / x) \cdot (1/x) - y^2 / x^2$.
- **Combine into a Single Fraction:**
- • $d^2y/dx^2 = 2y^3 / x^2 - y^2 / x^2 = (2y^3 - y^2) / x^2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Treats y^2 as having derivative 2y without applying the chain rule (dy/dx) or quotient rule.
✗ **(C)** Sign error in quotient/product rule, producing $+y^2$ instead of $-y^2$.
✗ **(D)** Omits the derivative of the denominator term (1/x).

Question 86 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 0.8; underestimate because f''(0) = 5 > 0

STEP-BY-STEP SOLUTION

- **Evaluate First Derivative at (0, 1):** $dy/dx|_{(0,1)} = 0 - 2(1) = -2$.
- **Write Tangent Line Equation:** $y - 1 = -2(x - 0) \Rightarrow y = 1 - 2x$.
- **Approximate f(0.1):** $f(0.1) \approx 1 - 2(0.1) = 1 - 0.2 = 0.8$.
- **Compute Second Derivative:**
- • $d^2y/dx^2 = d/dx [x - 2y] = 1 - 2(dy/dx)$.
- • At (0, 1): $d^2y/dx^2 = 1 - 2(-2) = 1 + 4 = 5 > 0$.
- **Determine Over/Underestimate:** Because $f''(0) = 5 > 0$, the curve is concave up at $x = 0$, so the tangent line lies below the curve. Thus 0.8 is an underestimate.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Signs error in evaluating f''(0): calculates $1 - 2(2) = -3$ or -5 and concludes overestimate.
✗ **(C)** Sign error in tangent slope: uses +2 instead of −2.
✗ **(D)** Combines wrong tangent slope with wrong concavity.

Question 87 Analysis

★★★★ MANDATORY

Correct Answer: (B) 600 grams

STEP-BY-STEP SOLUTION

- **Understand Equilibrium Concentration:** Because the inflow and outflow rates are identical (4 L/min), the total volume remains constant at 200 liters.
- **Steady-State Concentration:** As $t \to \infty$, the concentration of dye in the tank approaches the concentration of the incoming fluid: 3 grams/liter.
- **Compute Limiting Salt/Dye Amount:** Limiting amount = (Concentration) · (Volume) = (3 g/L) · (200 L) = 600 grams.
- **Differential Equation Verification:** $dS/dt = 12 - (4/200) S = 12 - 0.02 S$. Setting $dS/dt = 0$ gives $0.02 S = 12 \Rightarrow S = 12 / 0.02 = 600$ grams.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Equates limiting dye to volume: 200 grams.
✗ **(C)** Calculates (4)(200) = 800 grams.
✗ **(D)** Multiplies (3)(4)(100) erroneously.

Question 88 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) Points of the form (–1, y) where y > 2

STEP-BY-STEP SOLUTION

- **Find Critical Points (dy/dx = 0):** $dy/dx = (x + 1)(y - 2) = 0$ when $x = -1$ or $y = 2$.
- • If $y = 2$, $dy/dx = 0$ identically for all x, which is a horizontal constant line solution $y = 2$ (no strict local extremum).
- **Compute Second Derivative at Critical Points:**
- • $d^2y/dx^2 = d/dx\,[(x + 1)(y - 2)] = (1)(y - 2) + (x + 1)(dy/dx)$.
- • At $x = -1$: $d^2y/dx^2\,|_{(-1,\,y)} = (y - 2) + 0 = y - 2$.
- **Apply Second Derivative Test:**
- • For a relative minimum, we require $dy/dx = 0$ and $d^2y/dx^2 > 0$.
- • Therefore, $y - 2 > 0 \Rightarrow y > 2$.
- **Conclusion:** Relative minima occur at points $(-1, y)$ with $y > 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Along $y = 2$, the solution is constant, with no local extremum.
✗ **(C)** For $y < 2$, $d^2y/dx^2 < 0$, which corresponds to a relative maximum.
✗ **(D)** Confuses the conditions for x and y.

Question 89 Analysis

★★★★ MANDATORY

Correct Answer: (A) $e^{2y}\sin^2 x + e^y \cos x$

STEP-BY-STEP SOLUTION

- **Differentiate using Product Rule and Chain Rule:**
- • $d^2y/dx^2 = d/dx\,[e^y \cdot \sin x] = [d/dx(e^y)] \cdot \sin x + e^y \cdot [d/dx(\sin x)]$.
- **Apply Chain Rule to e^y:**
- • $d/dx(e^y) = e^y \cdot (dy/dx)$.
- **Substitute $dy/dx = e^y \sin x$:**
- • $d/dx(e^y) = e^y \cdot (e^y \sin x) = e^{2y} \sin x$.
- **Complete Expression:**
- • $d^2y/dx^2 = (e^{2y} \sin x) \cdot \sin x + e^y \cdot \cos x = e^{2y} \sin^2 x + e^y \cos x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets that $d/dx(e^y)$ produces an extra factor of $dy/dx = e^y \sin x$, missing e^{2y}.
✗ **(C)** Drops the second derivative product term.
✗ **(D)** Incorrectly groups e^y without applying the chain rule.

Question 90 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) A family of non-vertical lines passing through the point (0, 3)

STEP-BY-STEP SOLUTION

- **Separate Variables:** $(1 / (y - 3))\,dy = (1/x)\,dx$.
- **Integrate Both Sides:** $\int (1 / (y - 3))\,dy = \int (1/x)\,dx \Rightarrow \ln|y - 3| = \ln|x| + C_1 = \ln x + C_1$ (for $x > 0$).
- **Exponentiate:** $|y - 3| = e^{C_1} \cdot x \Rightarrow y - 3 = k\,x$ (where k is any real constant, including $k = 0$).
- **Isolate y:** $y = kx + 3$.
- **Interpret Geometrically:** The equation $y = kx + 3$ represents the family of all non-vertical straight lines with y-intercept (0, 3).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses linear separation with quadratic integration.
✗ **(B)** Assumes an exponential curve because of the presence of ln.
✗ **(D)** Confuses with $dy/dx = -x/(y - 3)$ which yields circles.

FRQ 1: Separation of Variables, Tangent Lines & Concavity Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Tangent Line and Approximation [2 Points]

- *Compute Slope:* $f'(2) = dy/dx\,|_{(2,\,1)} = (2-1)\cdot(1)^2 = (1)(1) = 1$.
- *Write Tangent Line:* Using point $(2, 1)$ and slope $m = 1$: $y - 1 = 1(x - 2) \Rightarrow y = x - 1$.
- *Approximate f(2.1):* $f(2.1) \approx 2.1 - 1 = 1.1$.

Part (b): Second Derivative and Concavity Justification [2 Points]

- *Differentiate $dy/dx = (x - 1)\,y^2$ using Product and Chain Rules:*
- $d^2y/dx^2 = d/dx(x - 1)\cdot y^2 + (x - 1)\cdot d/dx(y^2) = (1)\cdot y^2 + (x - 1)\cdot[2y\,(dy/dx)]$.
- *Evaluate at (2, 1) with $dy/dx = 1$:*
- $d^2y/dx^2\,|_{(2,\,1)} = (1)^2 + (2 - 1)\cdot[2(1)(1)] = 1 + 2 = 3$.
- *Justification:* Because $d^2y/dx^2 = 3 > 0$, the graph of f is concave up at and near $x = 2$. Consequently, the tangent line lies below the curve, so the approximation 1.1 is an **underestimate**.

Part (c): Particular Solution and Domain [5 Points]

- *Separate Variables:* $y^{-2}\,dy = (x - 1)\,dx$.
- *Integrate Both Sides:* $\int y^{-2}\,dy = \int (x - 1)\,dx \Rightarrow -1/y = x^2/2 - x + C$.
- *Apply Initial Condition f(2) = 1:* $-1/1 = 2^2/2 - 2 + C \Rightarrow -1 = 2 - 2 + C \Rightarrow C = -1$.
- *Substitute C:* $-1/y = x^2/2 - x - 1 = (x^2 - 2x - 2)\,/\,2$.
- *Solve for y:* $y = -2\,/\,(x^2 - 2x - 2) = 2\,/\,(2 + 2x - x^2)$.
- *Find Domain:* The denominator $2 + 2x - x^2 = 0$ when $x^2 - 2x - 2 = 0 \Rightarrow x = (2 \pm \sqrt{(4 + 8)})/2 = 1 \pm \sqrt{3}$.
- Since $1 - \sqrt{3} \approx -0.732$ and $1 + \sqrt{3} \approx 2.732$, the initial point $x = 2$ lies strictly between $1 - \sqrt{3}$ and $1 + \sqrt{3}$.
- *Domain of Validity:* $(1 - \sqrt{3},\, 1 + \sqrt{3})$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for tangent slope $f'(2) = 1$; 1 pt for tangent line $y = x - 1$ and approximation $f(2.1) \approx 1.1$	**2 pts**
Part (b)	1 pt for $d^2y/dx^2 = y^2 + 2(x - 1)\,y\,(dy/dx)$; 1 pt for evaluating to $3 > 0$ and concluding underestimate	**2 pts**
Part (c)	1 pt for separation of variables; 1 pt for antiderivatives; 1 pt for constant $C = -1/2$; 1 pt for solving for y; 1 pt for domain $(-\infty, 3)$	**5 pts**

FRQ 2: Euler's Method & Solution Analysis (BC Exclusive) Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Euler's Method Steps [3 Points]

- *Initial State:* $x_0 = 1, y_0 = 2$, step size $h = \Delta x = 0.5$.
- *Step 1 (from $x_0 = 1$ to $x_1 = 1.5$):*
- $dy/dx\,|_{(1,\,2)} = 2(1) - 2 = 0$.
- $y_1 = y_0 + (dy/dx) \cdot h = 2 + (0)(0.5) = 2.0$.
- *Step 2 (from $x_1 = 1.5$ to $x_2 = 2.0$):*
- $dy/dx\,|_{(1.5,\,2)} = 2(1.5) - 2.0 = 3 - 2 = 1.0$.
- $y_2 = y_1 + (dy/dx) \cdot h = 2.0 + (1.0)(0.5) = 2.5$.
- *Conclusion:* $f(2) \approx 2.5$.

Part (b): Second Derivative and Error Justification [2 Points]

- *Compute d^2y/dx^2:*
- $d^2y/dx^2 = d/dx\,[2x - y] = 2 - dy/dx = 2 - (2x - y) = 2 - 2x + y$.
- *Analyze Along the Solution Path:*
- At (1, 2): $d^2y/dx^2 = 2 - 2(1) + 2 = 2 > 0$.
- At (1.5, 2): $d^2y/dx^2 = 2 - 2(1.5) + 2 = 2 - 3 + 2 = 1 > 0$.
- *Justification:* Because $d^2y/dx^2 > 0$ on the interval [1, 2], the solution curve is concave up. Tangent segments lie below the curve at every step, so Euler's method produces an **underestimate**.

Part (c): Linear Solution Constants [2 Points]

- *Substitute $y = mx + b$ into dy/dx:* $dy/dx = m$.
- *Substitute into Differential Equation:* $m = 2x - (mx + b) = (2 - m)x - b$.
- *Equate Coefficients:*
- For the x-term: $2 - m = 0 \Rightarrow m = 2$.
- For the constant term: $m = -b \Rightarrow 2 = -b \Rightarrow b = -2$.
- *Linear Solution:* $y = 2x - 2$.

Part (d): Asymptotic Limit [2 Points]

- *Rewrite f(x):* Since $f(x) - (2x - 2) = g(x)$ with $g(x) \to 0$ as $x \to \infty$, $f(x) = 2x - 2 + g(x)$.
- *Evaluate Limit:* $\lim_{x\to\infty} [f(x)/x] = \lim_{x\to\infty} [(2x - 2 + g(x))/x] = \lim_{x\to\infty} [2 - 2/x + g(x)/x] = 2 - 0 + 0 = 2$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for step size $\Delta x = 0.5$; 1 pt for first step $f(1.5) \approx 2.0$; 1 pt for second step $f(2.0) \approx 2.5$	**3 pts**
Part (b)	1 pt for $d^2y/dx^2 = 2 - (2x - y) = 2 - 2x + y$; 1 pt for sign analysis ($d^2y/dx^2 > 0$) and concluding underestimate	**2 pts**
Part (c)	1 pt for setting $m = 2x - (mx + b)$; 1 pt for equating coefficients: $m = 2, b = -2$	**2 pts**
Part (d)	1 pt for relating $f(x)$ to $2x - 2$; 1 pt for final limit = 2	**2 pts**

FRQ 3: Logistic Population Dynamics (BC Exclusive) Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Intervals of Increase and Decrease [2 Points]

- *Analyze Sign of dP/dt:* $\frac{dP}{dt} = \frac{1}{200}P(1{,}000 - P)$.
- *Increasing:* $\frac{dP}{dt} > 0$ when $P > 0$ and $1{,}000 - P > 0 \Rightarrow$ **$0 < P < 1{,}000$**.
- *Decreasing:* $\frac{dP}{dt} < 0$ when $1{,}000 - P < 0 \Rightarrow$ **$P > 1{,}000$**.

Part (b): Maximum Growth Rate [2 Points]

- *Find Maximizing Population:* The rate $\frac{dP}{dt}$ is a quadratic function of P with roots $P = 0$ and $P = 1{,}000$. Its vertex occurs at $P = 1{,}000 / 2 =$ **500 elk**.
- *Calculate Maximum Rate:*
- $\left(\frac{dP}{dt}\right)_{max} = \frac{1}{200} \cdot (500) \cdot (1{,}000 - 500) = \frac{5}{2} \cdot 500 =$ **1,250 elk/year**.

Part (c): Long-term Limits and Interpretation [2 Points]

- *Limit of P(t):* Since $P(0) = 200 > 0$, $\lim_{t \to \infty} P(t) =$ **1,000 elk**. This is the carrying capacity of the habitat.
- *Limit of dP/dt:* As $P \to 1{,}000$, $\lim_{t \to \infty} \frac{dP}{dt} = \frac{1}{200}(1{,}000)(1{,}000 - 1{,}000) =$ **0 elk/year**. The population stabilizes, so growth ceases.

Part (d): Explicit Particular Solution [3 Points]

- *Separate Variables:* $\left[1 / (P(1{,}000 - P))\right] dP = \frac{1}{200}\, dt$.
- *Partial Fractions:* $1 / [P(1{,}000 - P)] = \frac{1}{1{,}000}\left[\frac{1}{P} + \frac{1}{1{,}000 - P}\right]$.
- Multiply by 1,000: $\left[\frac{1}{P} + \frac{1}{1{,}000 - P}\right] dP = 5\, dt$.
- *Integrate Both Sides:* $\ln|P| - \ln|1{,}000 - P| = 5t + C \Rightarrow \ln\left|\frac{P}{1{,}000 - P}\right| = 5t + C$.
- *Apply P(0) = 200:* $\ln|200 / 800| = \ln(1/4) = C \Rightarrow C = -\ln 4$.
- *Exponentiate:* $\frac{P}{1{,}000 - P} = \frac{1}{4} e^{5t}$.
- *Invert and Solve:* $\frac{(1{,}000 - P)}{P} = 4e^{-5t} \Rightarrow \frac{1{,}000}{P} - 1 = 4e^{-5t} \Rightarrow \frac{1{,}000}{P} = 1 + 4e^{-5t}$.
- *Final Solution:* **$P(t) = \frac{1{,}000}{1 + 4e^{-5t}}$**.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for increasing when $0 < P < 1{,}000$; 1 pt for decreasing when $P > 1{,}000$ with dP/dt sign analysis	**2 pts**
Part (b)	1 pt for P = 500 elk; 1 pt for maximum growth rate = 1,250 elk/year	**2 pts**
Part (c)	1 pt for $\lim P(t) = 1{,}000$ (carrying capacity); 1 pt for $\lim \frac{dP}{dt} = 0$ (growth ceases)	**2 pts**
Part (d)	1 pt for partial fractions separation; 1 pt for constant evaluation; 1 pt for $P(t) = 1{,}000 / (1 + 4e^{-5t})$	**3 pts**

FRQ 4: Slope Fields & Particular Solutions with Radical Branches Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Geometric Reasoning along Coordinate Lines [2 Points]

- *Along $x = -2$:* $dy/dx = (-2 + 2) / y = 0 / y = 0$ (for all $y \neq 0$). The slope field consists of **horizontal line segments** with slope 0.
- *Along $y = 0$:* $dy/dx = (x + 2) / 0$, which is undefined. The tangents are **vertical** along the x-axis.

Part (b): Tangent Line Approximation [2 Points]

- *Find Tangent Slope:* $f'(0) = dy/dx|_{(0, -2)} = (0 + 2) / (-2) = 2 / (-2) = -1$.
- *Write Tangent Line:* Using point $(0, -2)$ and slope $m = -1$: $y - (-2) = -1(x - 0) \Rightarrow y = -x - 2$.
- *Approximate $f(0.2)$:* $f(0.2) \approx -0.2 - 2 =$ **-2.2**.

Part (c): Particular Solution and Domain [5 Points]

- *Separate Variables:* $y\,dy = (x + 2)\,dx$.
- *Integrate Both Sides:* $\int y\,dy = \int (x + 2)\,dx \Rightarrow y^2/2 = (x + 2)^2/2 + C_1 \Rightarrow y^2 = (x + 2)^2 + C$.
- *Apply Initial Condition $f(0) = -2$:* $(-2)^2 = (0 + 2)^2 + C \Rightarrow 4 = 4 + C \Rightarrow C = 0$.
- *Alternative Standard Form:* $y^2/2 = x^2/2 + 2x + C_2 \Rightarrow y^2 = x^2 + 4x + 2C_2$.
- At $(0, -2)$: $(-2)^2 = 0 + 0 + 2C_2 \Rightarrow 4 = 2C_2$. Thus $y^2 = x^2 + 4x + 4 = (x + 2)^2$.
- *Notice:* If $C = 0$, $y^2 = (x + 2)^2$. Since $y(0) = -2$, $y = -|x + 2| = -(x + 2)$ for $x > -2$.
- *Wait, verify:* $dy/dx = -1$, while $(x+2)/y = (x+2)/[-(x+2)] = -1$ for $x > -2$! Indeed, $dy/dx = -1$ everywhere!
- *Domain of Validity:* For dy/dx to exist, $y \neq 0 \Rightarrow x + 2 \neq 0 \Rightarrow x \neq -2$. Since $x_0 = 0 > -2$, the largest open interval is **$(-2, \infty)$**.
- *Explicit Formula:* **$y = -(x + 2)$ on $(-2, \infty)$**.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for horizontal tangent slopes (slope = 0) along $x = -2$; 1 pt for vertical tangents (undefined slope) along $y = 0$	**2 pts**
Part (b)	1 pt for tangent line $y = -x - 2$; 1 pt for approximation $f(0.2) \approx -2.2$	**2 pts**
Part (c)	1 pt for $y\,dy = (x + 2)\,dx$; 1 pt for integration; 1 pt for constant $C = 4$; 1 pt for negative square root branch; 1 pt for open domain $(-2 - 2\sqrt{2}, -2 + 2\sqrt{2})$	**5 pts**

FRQ 5: Mixing Tank Differential Equation & Steady-State Official Model Solution

★★★★ MANDATORY

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Differential Equation Formulation [2 Points]

- *Compute Inflow Rate:* (2 lbs/gal) · (3 gal/min) = 6 lbs/min.
- *Compute Outflow Rate:* Volume is constant at 100 gallons. Concentration = S(t) / 100 lbs/gal.
- Rate_out = (S(t) / 100) · (3 gal/min) = 0.03 S lbs/min.
- *Differential Equation:* **$dS/dt = 6 - 0.03\,S$** with initial condition **$S(0) = 50$**.

Part (b): Separation of Variables Solution [4 Points]

- *Rewrite Differential Equation:* $dS/dt = -0.03\,(S - 200)$.
- *Separate Variables:* $(1 / (S - 200))\,dS = -0.03\,dt$.
- *Integrate Both Sides:* $\int (1 / (S - 200))\,dS = \int -0.03\,dt \Rightarrow \ln|S - 200| = -0.03\,t + C_1$.
- *Exponentiate:* $|S - 200| = e^{C_1} \cdot e^{-0.03\,t} \Rightarrow S - 200 = A\,e^{-0.03\,t}$.
- *Apply S(0) = 50:* $50 - 200 = A\,e^{0} \Rightarrow A = -150$.
- *Solve for S(t):* **$S(t) = 200 - 150\,e^{-0.03\,t}$**.

Part (c): Value at t = 10 Minutes [1 Point]

- $S(10) = 200 - 150\,e^{-0.03 \cdot 10} =$ **$200 - 150\,e^{-0.3}$ pounds**.

Part (d): Limiting Value and Physical Meaning [2 Points]

- *Evaluate Limit:* As $t \to \infty$, $e^{-0.03\,t} \to 0$, so **$\lim_{t \to \infty} S(t) = 200$ pounds**.
- *Physical Explanation:* Over a long time, the tank fluid is completely replaced by incoming brine with concentration 2 lbs/gal. The total volume remains 100 gallons, yielding a steady-state salt mass of (2 lbs/gal) · (100 gal) = 200 pounds.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for $dS/dt = 6 - 0.03\,S$; 1 pt for initial condition $S(0) = 50$	**2 pts**
Part (b)	1 pt for separation of variables; 1 pt for integration; 1 pt for constant $C = -150$; 1 pt for $S(t) = 200 - 150\,e^{-0.03\,t}$	**4 pts**
Part (c)	1 pt for $S(10) = 200 - 150\,e^{-0.3}$ pounds	**1 pt**
Part (d)	1 pt for limit = 200 pounds; 1 pt for explanation that 2 lbs/gal × 100 gal = 200 lbs	**2 pts**

FRQ 6: Euler's Method & Taylor Polynomial Comparison (BC Exclusive) Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Euler's Method [3 Points]

- *Initial State:* $x_0 = 0, y_0 = 1, h = \Delta x = 0.2$.
- *Step 1 ($x_0 = 0$ to $x_1 = 0.2$):*
- $dy/dx\,|_{(0,\,1)} = 0^2 + 1 = 1$.
- $y_1 = y_0 + (dy/dx) \cdot h = 1 + (1)(0.2) = 1.2$.
- *Step 2 ($x_1 = 0.2$ to $x_2 = 0.4$):*
- $dy/dx\,|_{(0.2,\,1.2)} = (0.2)^2 + 1.2 = 0.04 + 1.2 = 1.24$.
- $y_2 = y_1 + (dy/dx) \cdot h = 1.2 + (1.24)(0.2) = 1.2 + 0.248 = \mathbf{1.448}$.

Part (b): Second Derivative [2 Points]

- *Differentiate with Respect to x:* $d^2y/dx^2 = d/dx\,[x^2 + y] = 2x + dy/dx$.
- *Substitute $dy/dx = x^2 + y$:* $d^2y/dx^2 = 2x + x^2 + y$.
- *Evaluate at (0, 1):* $d^2y/dx^2\,|_{(0,\,1)} = 2(0) + 0^2 + 1 = \mathbf{1}$.

Part (c): Error Justification [2 Points]

- For all $x \in [0, 0.4]$ and $y \geq 1$, $x \geq 0$ and $y > 0$, so $d^2y/dx^2 = 2x + x^2 + y > 0$.
- Because the function is concave up throughout this interval, tangent segments lie below the curve at every stage.
- Therefore, the Euler approximation 1.448 is **less than** the actual value of f(0.4).

Part (d): Second-Degree Taylor Polynomial [2 Points]

- *Compute Coefficients at $x = 0$:* $f(0) = 1, f'(0) = 1, f''(0) = 1$.
- *Formula:* $P_2(x) = f(0) + f'(0)\,x + [f''(0) / 2!]\,x^2 = \mathbf{1 + x + x^2/2}$.
- *Approximate f(0.4):* $P_2(0.4) = 1 + 0.4 + (0.4)^2/2 = 1 + 0.4 + 0.16/2 = 1 + 0.4 + 0.08 = \mathbf{1.48}$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for step size 0.2; 1 pt for $f(0.2) \approx 1.2$; 1 pt for $f(0.4) \approx 1.448$	**3 pts**
Part (b)	1 pt for $d^2y/dx^2 = 2x + dy/dx = 2x + x^2 + y$; 1 pt for value at (0, 1) = 1	**2 pts**
Part (c)	1 pt for stating $d^2y/dx^2 > 0$; 1 pt for concluding Euler's method gives an underestimate (less than actual)	**2 pts**
Part (d)	1 pt for $P_2(x) = 1 + x + x^2/2$; 1 pt for $P_2(0.4) = 1 + 0.4 + 0.08 = 1.48$	**2 pts**

FRQ 7: Newton's Law of Cooling & Rate Analysis Official Model Solution

★★★★ MANDATORY

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Separation of Variables Derivation [3 Points]

- *Separate Variables:* $(1 / (T - 65))\, dT = -k\, dt$.
- *Integrate Both Sides:* $\int (1 / (T - 65))\, dT = \int -k\, dt \Rightarrow \ln|T - 65| = -kt + C_1$.
- *Exponentiate:* $|T - 65| = e^{C_1} \cdot e^{-kt} \Rightarrow T - 65 = A\, e^{-kt}$.
- *Apply Initial Condition T(0) = 185:* $185 - 65 = A\, e^{0} \Rightarrow A = 120$.
- *Result:* **$T(t) = 65 + 120\, e^{-kt}$**.

Part (b): Exact Cooling Constant k [2 Points]

- *Apply T(10) = 125:* $125 = 65 + 120\, e^{-10k}$.
- *Isolate Exponential:* $60 = 120\, e^{-10k} \Rightarrow e^{-10k} = 60 / 120 = 1/2$.
- *Take Natural Logarithm:* $-10k = \ln(1/2) = -\ln 2 \Rightarrow$ **$k = (\ln 2) / 10$**.

Part (c): Time to Reach 85°F [2 Points]

- *Set T(t) = 85:* $85 = 65 + 120\, e^{-kt} \Rightarrow 20 = 120\, e^{-kt}$.
- *Isolate Exponential:* $e^{-kt} = 20 / 120 = 1/6$.
- *Solve for t:* $-kt = \ln(1/6) = -\ln 6 \Rightarrow t = (\ln 6) / k$.
- *Substitute k:* $t = (\ln 6) / [(\ln 2) / 10] =$ **$10\, (\ln 6) / (\ln 2)$ minutes**.

Part (d): Rate of Cooling at t = 10 Minutes [2 Points]

- *Evaluate Rate Equation:* $dT/dt = -k\, (T - 65)$.
- At $t = 10$, $T = 125$, so $dT/dt\,|_{(10)} = -[(\ln 2) / 10] \cdot (125 - 65) = -[(\ln 2) / 10] \cdot 60 =$ **$-6 \ln 2$ °F/min**.
- *Interpretation:* At time $t = 10$ minutes, the metal bar's temperature is decreasing at a rate of $6 \ln 2 \approx 4.16$ degrees Fahrenheit per minute.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for separation of variables; 1 pt for integration; 1 pt for applying $T(0) = 185$ to get $C = 120$	**3 pts**
Part (b)	1 pt for $60 = 120\, e^{-10k}$; 1 pt for $k = (\ln 2) / 10$	**2 pts**
Part (c)	1 pt for setting $85 = 65 + 120\, e^{-kt}$; 1 pt for $t = (\ln 6) / k = 10\, (\ln 6) / (\ln 2)$	**2 pts**
Part (d)	1 pt for $dT/dt = -6 \ln 2$; 1 pt for interpretation: bar is cooling at a rate of $6 \ln 2$ °F/min at $t = 10$	**2 pts**

FRQ 8: Separation of Variables with Inverse Trigonometric Forms Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Tangent Slope at x = 0 [1 Point]

- $dy/dx\,|_{(0,\,1)} = (1 + 1^2) \cdot e^0 = (2)(1) = \mathbf{2}$.

Part (b): Second Derivative and Concavity [2 Points]

- *Differentiate using Product Rule:*
- $d^2y/dx^2 = d/dx(1 + y^2) \cdot e^x + (1 + y^2) \cdot d/dx(e^x) = [2y\,(dy/dx)]\,e^x + (1 + y^2)\,e^x$.
- *Evaluate at (0, 1) with dy/dx = 2:*
- $d^2y/dx^2\,|_{(0,\,1)} = [2(1)(2)] \cdot (1) + (1 + 1^2) \cdot (1) = 4 + 2 = \mathbf{6}$.
- *Conclusion:* Because $d^2y/dx^2 = 6 > 0$, the graph is **concave up** at $x = 0$.

Part (c): Particular Solution [4 Points]

- *Separate Variables:* $[1 / (1 + y^2)]\,dy = e^x\,dx$.
- *Integrate Both Sides:* $\int [1 / (1 + y^2)]\,dy = \int e^x\,dx \Rightarrow \arctan(y) = e^x + C$.
- *Apply Initial Condition f(0) = 1:* $\arctan(1) = e^0 + C \Rightarrow \pi/4 = 1 + C \Rightarrow C = \pi/4 - 1$.
- *Solve for y:* $\arctan(y) = e^x + \pi/4 - 1 \Rightarrow \mathbf{y = \tan(e^x + \pi/4 - 1)}$.

Part (d): Domain of Definition [2 Points]

- For tan(u) to be continuous on an interval containing the initial point $u(0) = \pi/4$, we must have $-\pi/2 < u < \pi/2$.
- $-\pi/2 < e^x + \pi/4 - 1 < \pi/2$.
- *Lower Bound:* $e^x > 1 - 3\pi/4 \approx 1 - 2.356 = -1.356$. Since $e^x > 0$ for all real x, this condition holds for all $x \in \mathbb{R}$.
- *Upper Bound:* $e^x < 1 + \pi/4 \Rightarrow x < \ln(1 + \pi/4)$.
- *Domain:* $\mathbf{(-\infty,\ \ln(1 + \pi/4))}$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for slope = 2	**1 pt**
Part (b)	1 pt for $d^2y/dx^2 = 2y\,(dy/dx)\,e^x + (1 + y^2)\,e^x$; 1 pt for evaluating to $6 > 0$ and concluding concave up	**2 pts**
Part (c)	1 pt for separation of variables; 1 pt for $\arctan y = e^x + C$; 1 pt for $C = \pi/4 - 1$; 1 pt for $y = \tan(e^x + \pi/4 - 1)$	**4 pts**
Part (d)	1 pt for setting $-\pi/2 < e^x + \pi/4 - 1 < \pi/2$; 1 pt for domain $(-\infty, \ln(\pi/4 + 1))$	**2 pts**

FRQ 9: Comparative Population Growth Models (BC Exclusive) Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Model A Solution and Doubling Time [2 Points]

- *Explicit Solution:* $dy/dt = 0.2\,y \Rightarrow$ **$y(t) = 50\,e^{0.2\,t}$**.
- *Doubling Time:* $100 = 50\,e^{0.2\,t} \Rightarrow e^{0.2\,t} = 2 \Rightarrow 0.2\,t = \ln 2 \Rightarrow$ **$t = (\ln 2) / 0.2 = 5 \ln 2$ days**.

Part (b): Model B Logistic Characteristics [3 Points]

- *Carrying Capacity:* Comparing with $dP/dt = kP(1 - P/M)$, **$M = 500$ organisms**.
- *Fastest Growth Population:* $P = M / 2 = 500 / 2 =$ **250 organisms**.
- *Maximum Growth Rate:* $(dP/dt)_{max} = 0.2 \cdot (250) \cdot (1 - 250/500) = 50 \cdot (0.5) =$ **25 organisms/day**.

Part (c): Ratio Limit of Rates [2 Points]

- As $t \to \infty$, $P(t) \to 500$, so $dP/dt = 0.2\,P\,(1 - P/500) \to 0.2(500)(1 - 1) = 0$.
- As $t \to \infty$, $y(t) = 50\,e^{0.2\,t} \to \infty$, so $dy/dt = 0.2\,y \to \infty$.
- *Evaluate Limit:* $\lim_{t \to \infty} [\,(dP/dt) / (dy/dt)\,] = 0 / \infty =$ **0**.

Part (d): Biological Comparison [2 Points]

- *Asymptotic Contrast:* y(t) grows exponentially without bound, whereas P(t) is bounded above by 500 organisms.
- *Realism Justification:* **Model B (Logistic)** is far more realistic in any natural ecosystem because food, space, and oxygen are finite, imposing an environmental carrying capacity.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for $y(t) = 50\,e^{0.2\,t}$; 1 pt for doubling time $t = (\ln 2) / 0.2 = 5 \ln 2$ days	**2 pts**
Part (b)	1 pt for carrying capacity $M = 500$; 1 pt for fastest growth at $P = 250$; 1 pt for max rate = 25 organisms/day	**3 pts**
Part (c)	1 pt for recognizing $dP/dt \to 0$ and $dy/dt \to \infty$; 1 pt for concluding limit = 0	**2 pts**
Part (d)	1 pt for stating y(t) grows without bound while P(t) approaches 500; 1 pt for explaining Model B is realistic due to carrying capacity constraints	**2 pts**

FRQ 10: Differential Equations Synthesis Masterpiece Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Horizontal Tangent Locations [1 Point]

- Set $dy/dx = 0$: $(x\,y) / 2 = 0$.
- Since the problem states $y > 0$, we must have $x = 0$.
- *Conclusion:* Horizontal tangents occur at all points **(0, y) where y > 0** (the positive y-axis).

Part (b): Euler's Method [2 Points]

- *Parameters:* $x_0 = 0$, $y_0 = 3$, $h = \Delta x = 0.5$.
- *Step 1 ($x_0 = 0$ to $x_1 = 0.5$):*
- $dy/dx\,|_{(0,\,3)} = (0 \cdot 3) / 2 = 0$.
- $y_1 = 3 + 0 \cdot (0.5) = 3.0$.
- *Step 2 ($x_1 = 0.5$ to $x_2 = 1.0$):*
- $dy/dx\,|_{(0.5,\,3)} = (0.5 \cdot 3.0) / 2 = 1.5 / 2 = 0.75$.
- $y_2 = 3.0 + (0.75) \cdot (0.5) = 3.0 + 0.375 =$ **3.375**.

Part (c): Second Derivative and Relative Extremum [2 Points]

- *Compute d^2y/dx^2:*
- $d^2y/dx^2 = d/dx\,[(1/2)\,x\,y] = (1/2)\,y + (1/2)\,x\,(dy/dx)$.
- Substitute $dy/dx = (x\,y) / 2$: $d^2y/dx^2 = y/2 + (x^2\,y) / 4$.
- *Evaluate at (0, 3):* $d^2y/dx^2\,|_{(0,\,3)} = 3/2 + 0 =$ **3/2 > 0**.
- *Second Derivative Test:* Since $dy/dx\,|_{(0,\,3)} = 0$ and $d^2y/dx^2\,|_{(0,\,3)} = 3/2 > 0$, the graph of f has a **relative minimum** at $x = 0$.

Part (d): Exact Particular Solution & Comparison [4 Points]

- *Separate Variables:* $(1/y)\,dy = (x / 2)\,dx$.
- *Integrate Both Sides:* $\int (1/y)\,dy = \int (x / 2)\,dx \Rightarrow \ln|y| = x^2 / 4 + C$.
- *Apply f(0) = 3:* $\ln(3) = 0 / 4 + C \Rightarrow C = \ln 3$.
- *Exponentiate:* $y = e^{x^2/4 + \ln 3} = e^{\ln 3} \cdot e^{x^2/4} =$ **$3\,e^{x^2/4}$**.
- *Compute Exact Value f(1):* $f(1) = 3\,e^{1/4} = 3\,e^{0.25} \approx 3(1.2840) = 3.852$.
- *Comparison:* The Euler approximation 3.375 is an underestimate of the exact value $3\,e^{1/4} \approx 3.852$, which is consistent with the curve being concave up ($d^2y/dx^2 > 0$).

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	1 pt for $x = 0$ (all points on the y-axis with $y > 0$)	**1 pt**
Part (b)	1 pt for first step $f(0.5) \approx 3.0$; 1 pt for second step $f(1.0) \approx 3.375$	**2 pts**
Part (c)	1 pt for $d^2y/dx^2 = y/2 + (x^2y)/4$; 1 pt for evaluating to $3/2 > 0$ and concluding relative minimum by Second Derivative Test	**2 pts**
Part (d)	1 pt for separation of variables; 1 pt for integration $\ln y = x^2/4 + C$; 1 pt for $f(x) = 3\,e^{x^2/4}$; 1 pt for exact $f(1) = 3\,e^{1/4} \approx 3.852 > 3.375$	**4 pts**

Essential Unit 7 Rules

Core formulas, differential equation models, and numerical methods required for AP Calculus BC.

- **Separation of Variables:** $dy/dx = g(x)\,h(y) \Rightarrow (1/h(y))\,dy = g(x)\,dx$. Integrate both sides and immediately append the constant of integration $+C$ before exponentiating or isolating y.
- **Domain of a Solution:** The solution to an initial value problem is defined on the *largest open interval containing the initial point* x_0 on which both the function and its derivative exist and are continuous.
- **Exponential Growth & Decay:** $dy/dt = ky \Rightarrow y(t) = y_0\,e^{(kt)}$. Doubling time / half-life: $t = (\ln 2) / |k|$.
- **Newton's Law of Cooling:** $dT/dt = -k(T - T_m) \Rightarrow T(t) = T_m + (T_0 - T_m)\,e^{(-kt)}$, where T_m is the ambient temperature.
- **Euler's Method (BC Topic):** $x_{n+1} = x_n + \Delta x$, $y_{n+1} = y_n + f'(x_n, y_n) \cdot \Delta x$.
- **Euler / Tangent Line Error & Concavity:** • If $d^2y/dx^2 > 0$ (concave up), tangent segments lie below the curve $\Rightarrow$ **Underestimate**. • If $d^2y/dx^2 < 0$ (concave down), tangent segments lie above the curve $\Rightarrow$ **Overestimate**.
- **Logistic Growth Model (BC Topic):** $dP/dt = k\,P\,(1 - P/M)$ or $dP/dt = k\,P\,(M - P)$. • Carrying capacity: $M = \lim(t\to\infty)\,P(t)$. • Point of fastest population growth: $P = M/2$ (inflection point of logistic curve). • Maximum rate of growth: $(dP/dt)_{max} = k\,M/4$. • Explicit solution: $P(t) = M / (1 + A\,e^{(-kt)})$, where $A = (M - P_0)/P_0$.
- **Mixing Tank Models:** $dS/dt = \text{Rate_in} - \text{Rate_out} = (c_in \cdot r_in) - (S(t)/V(t) \cdot r_out)$.

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Thank You, Students!

Mastery Comes Through Deliberate Practice

Congratulations on completing this comprehensive 100-Question Priority-Starred Master Repository for Unit 7! Differential equations are at the heart of calculus and physical modeling. Review your steps, reflect on the distractor rationales, and approach your AP Calculus BC examination with absolute mastery and confidence!

TARGET SCORE: 5 · AP CALCULUS BC